Professor Messer

Professor Messer's CompTIA A+ CORE 1 220-1201 Course Notes

James "Professor" Messer

Professor Messer's CompTIA 220-1201 Core 1 A+ Course Notes

James "Professor" Messer

http://www.ProfessorMesser.com

Professor Messer's CompTIA 220-1201 Core 1 A+ Course Notes
Written by James "Professor" Messer
Copyright © 2025 by Messer Studios, LLC
http://www.ProfessorMesser.com
All rights reserved. No part of this book may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording, or by any information storage and retrieval system, without written permission from the publisher.

First Edition: March 2025
This is version 1.50

Trademark Acknowledgments
All product names and trademarks are the property of their respective owners, and are in no way associated or affiliated with Messer Studios LLC.

"Professor Messer" is a registered trademark of Messer Studios LLC.

"CompTIA" and "A+" are registered trademarks of CompTIA, Inc.

Warning and Disclaimer
This book is designed to provide information about the CompTIA 220-1201 A+ certification exam. However, there may be typographical and/or content errors. Therefore, this book should serve only as a general guide and not as the ultimate source of subject information. The author shall have no liability or responsibility to any person or entity regarding any loss or damage incurred, or alleged to have incurred, directly or indirectly, by the information contained in this book.

Contents

1.0 - Mobile Devices 1
1.1 - Laptop Hardware 1
1.2 - Connecting Mobile Devices 2
1.2 - Mobile Device Accessories 3
1.3 - Mobile Device Networks 4
1.3 - Mobile Device Management 5

2.0 - Networking 6
2.1 - Introduction to IP 6
2.1 - Common Ports 7
2.2 - Wireless Network Technologies 8
2.3 - Network Services 9
2.4 - DNS Configuration 11
2.4 - DHCP 13
2.4 - VLANs and VPNs 14
2.5 - Network Devices 15
2.6 - IPv4 and IPv6 17
2.6 - Assigning IP Addresses 18
2.7 - Internet Connection Types 18
2.7 - Network Types 19
2.8 - Network Tools 20

3.0 - Hardware 21
3.2 - Network Cables 22
3.2 - 568A and 568B Colors 24
3.2 - Optical Fiber 24
3.2 - Peripheral Cables 25
3.2 - Video Cables 26
3.2 - Storage Cables 27
3.2 - Adapters and Converters 27
3.2 - Copper Connectors 28
3.2 - Fiber Connectors 29
3.3 - An Overview of Memory 30
3.3 - Memory Technologies 31
3.4 - Storage Devices 32
3.4 - RAID 33
3.5 - Motherboard Form Factors 34
3.5 - Motherboard Expansion Slots 35
3.5 - Motherboard Connections 35
3.5 - Motherboard Compatibility 35

3.5 - The BIOS 36
3.5 - BIOS Settings 36
3.5 - HSM and TPM 37
3.5 - CPU Features 38
3.5 - Expansion Cards 38
3.5 - Cooling 39
3.6 - Computer Power 40
3.7 - Multifunction Devices 41
3.8 - Laser Printer Maintenance 43
3.8 - Inkjet Printers 44
3.8 - Inkjet Printer Maintenance 44
3.8 - Thermal Printers 44
3.8 - Thermal Printer Maintenance 45
3.8 - Impact Printers 45
3.8 - Impact Printer Maintenance 45

4.0 - Virtualization and Cloud Computing 46
4.1 - Virtualization Concepts 46
4.1 - Virtualization Services 46
4.2 - Cloud Models 47
4.2 - Cloud Characteristics 48

5.0 - Hardware and Network Troubleshooting 48
5.1 - Troubleshooting Hardware 48
5.2 - Troubleshooting Storage Devices 50
5.3 - Troubleshooting Display Issues 52
5.4 - Troubleshooting Mobile Devices 53
5.5 - Troubleshooting Networks 55
5.6 - Troubleshooting Printers 56

Introduction

The CompTIA A+ certification requires a broad set of knowledge, and it covers more topics than many industry certifications. It's no surprise that the A+ certification has become one of the most sought-after industry certifications by both aspiring technologists and employers.

I hope this book helps you with your "last mile" of studies before taking your exam. There's a lot to remember, and perhaps some of the information in this book will help jog your memory while you're sitting in the exam room. Best of luck with your studies!

- Professor Messer

The CompTIA A+ Certification

CompTIA's A+ certification is considered to be the starting point for information technology professionals. Earning the A+ certification requires the completion of two exams and covers a broad range of technology topics. After earning the CompTIA A+ certification, an A+ certified professional will have an understanding of computer hardware, mobile devices, networking, operating systems, security techniques, and much more.

The current series of the A+ certification is based on the successful completion of the 220-1201 and the 220-1202 exams. You must pass both exams to earn your CompTIA A+ certification. This book provides a set of notes for the 220-1201 Core 1 exam.

The 220-1201 Core 1 exam

Much of the 220-1201 exam is about hardware, but that's only about a third of the overall exam content. You'll also need to know about networking, mobile devices, and how to troubleshoot all of these technologies.

Here's the breakdown of each domain on the 220-1201 exam:

Domain 1.0 - Mobile Devices - 13%
Domain 2.0 - Networking - 23%
Domain 3.0 - Hardware - 25%
Domain 4.0 - Virtualization and Cloud Computing - 11%
Domain 5.0 - Hardware and Network Troubleshooting - 28%

Study Tips

Exam Preparation

- Download the exam objectives, and use them as a master checklist: https://www.ProfessorMesser.com/objectives
- Use as many training materials as possible. Books, videos, and Q&A guides can all provide a different perspective of the same information.
- It's useful to have as much hands-on as possible, especially with network troubleshooting and operating system command prompts.

Taking the Exam

- Use your time wisely. You've got 90 minutes to get through everything.
- Choose your exam location carefully. Some sites are better than others.
- Get there early. Don't stress the journey.
- Manage your time wisely. You've got 90 minutes to get through everything.
- Wrong answers aren't counted against you. Don't leave any blanks!
- Mark difficult questions and come back later. You can answer the questions in any order.

Professor
Messer

Professor Messer's
CompTIA A+
220-1201 Course Notes

http://www.ProfessorMesser.com

1.1 - Laptop Hardware

Laptop hardware
- Engineered to precise specifications
  - A challenging repair
- Some laptops are easier to fix than others
  - An ongoing learning process
- Understand the process
  - The details will vary between laptop manufacturers

Laptop batteries
- The power source when unplugged
  - May be easily replaced or require a tear down
  - Five minutes vs. one hour
- Lithium-Ion (Li-ion) and Lithium-Ion polymer (LiPo) are common
  - No "memory effect"
  - Charging the battery does not diminish capacity
- Different form factor for each laptop
  - Battery types and styles can vary
  - Battery technology is changing constantly

Laptop keyboard
- The most-used component of the laptop
  - Can be easy to replace
  - A few screws and a single ribbon cable
  - May not always be this simple
- Or connect an external USB keyboard
  - Not very portable, but works in a pinch

Laptop keys
- Some repairs might require the removal or replacement of a key cap
  - This can be a delicate procedure
- Check with the manufacturer's instructions
  - It's very easy to accidentally break the key cap or the components underneath

Laptop memory
- Small Outline Dual In-line Memory Module (SO-DIMM)
- Memory used in laptops and mobile devices
  - Often easy to install and replace
- Some laptop memory is soldered to the system board
  - No upgrade available
  - Requires a full system board replacement

Laptop storage
- Magnetic disk
  - Traditional spinning drive platters
  - 2.5 form factors (3.5 inch for desktops)
- SSD (Solid-state drive)
  - All memory, no moving parts
  - Silent, fast access time, less latency
  - 2.5 inch form factors
- M.2
  - Smaller form factor
  - No SATA data or power cables
  - Easy to install and replace

Replacing laptop storage
- All internal
  - Open a cover on the back
  - Open the entire laptop
- Can be very modular
  - Two screws and the drive slides out
- M.2 drives are even easier
  - One screw - similar to RAM installation

Migrating from HDD to SSD
- An impressive upgrade
  - Move from spinning mechanical drive to solid state memory
  - It's almost like getting a new laptop
- Install an OS on the SSD
  - Move user documents between drives
  - Install any required applications
  - Can be time consuming
- Image/clone the HDD
  - No OS installation required
  - Move everything from one to the other
- Imaging software needed
  - Sometimes included with the SSD
  - Many commercial and open source options
- Create an image file
  - One drive at a time
- Drive-to-drive image
  - Image directly from one drive to the other

© 2025 Messer Studios, LLC

1.1 - Laptop Hardware (continued)

802.11 wireless and Bluetooth
- Wireless network connectivity
  - Connect without wires
- 802.11
  - Local area network (LAN)
  - High speed, Internet access
- Bluetooth
  - Personal area network (PAN) - Short range
  - Connect peripherals and other nearby devices

Biometrics
- Sign in or unlock your laptop with a fingerprint reader or face recognition
  - Something you are
- Requires additional configuration in the OS
  - Hardware required for most options
- Relatively secure
  - Faces and fingerprints are quite unique

Near-field communication (NFC)
- Short-distance networking
  - 4 centimeters or less
  - Data transfers or authentication
- Common on mobile phones and smart watches
  - Payment method on your wrist
- Use it for authentication without typing a password
  - Hospital workstations, warehouses, manufacturing

Wi-Fi antennas
- Multiple antennas
  - WiFi main and aux
  - Bluetooth
- Antenna wires wrap around the laptop screen
  - It's up high!

Microphone
- Built-in to the laptop display
  - Useful for video calls
- Not good for non-casual use
  - Analog or USB microphones can help

1.2 - Connecting Mobile Devices

Connecting mobile devices
- Phones and tablets
  - Many different ways to connect
- Wired and wireless connectivity
  - Standards and options change through the years
- Used for synchronization, connectivity, backup, and identification
  - Sometimes simultaneously

USB
- Universal Serial Bus
  - High-speed wired communication
  - Used for charging and data transfers
- Micro-USB
  - A smaller USB connection
  - Common now worldwide
- Older devices may use Mini-USB
  - Slightly larger

USB-C
- 24-pin double-sided USB connector
  - Used for both hosts and devices
- Acts as a USB 2.0/3.0/3.1/4 connection
  - Different signals can use the same connector
- Can transmit other signals
  - DisplayPort
  - HDMI
  - Thunderbolt

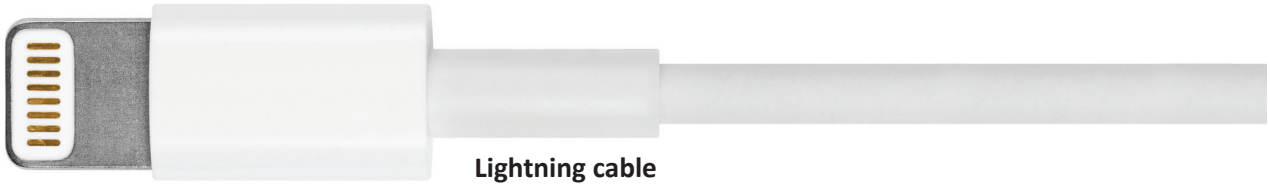
Lightning
- Apple proprietary
  - 8-pin digital signals
  - iPhone and iPad devices
- Some advantages over Micro-USB
  - Higher power output for phones and tablets
  - Can be inserted either way
  - Simpler design

NFC
- Near Field Communication
  - Send small amounts of data wirelessly over a limited area
- Built into your phone
  - Payment systems, transportation, in-person information exchange
- Access token, identity "card"
  - Short range with encryption support

1.2 - Connecting Mobile Devices (continued)

Bluetooth
- High speed communication over short distances
  - PAN (Personal Area Network)
- Connects our mobile devices
  - Smartphones
  - Tethering
  - Headsets and headphones
  - Health monitors
  - Automobile and phone integration
  - Smartwatches
  - External speakers

Hotspot/Tethering
- Turn your phone into a WiFi hotspot
  - Your own personal wireless router
  - Extend the cellular data network to all of your devices
- Dependent on phone type and provider
  - May require additional charges and data costs



Lightning cable

1.2 - Mobile Device Accessories

Stylus
- Digital stylus
  - A more advanced writing tool
- The stylus communicates directly to the device
  - Pressure sensitivity, programmable buttons, etc.
- Must be compatible with the tablet
  - e.g., Apple iPad uses an Apple Pencil

Headsets
- Hands-free audio - Earphones and microphone
- Wired
  - USB connections are common on laptops
  - Connects to 3.5 mm TRRS (tip-ring-ring-sleeve) connector
  - Analog audio jack
  - iPhone can use Lightning port
- Wireless - Bluetooth headsets

Speakers
- Mobile audio - Battery powered
- Wireless connection - Bluetooth link
- Stereo sound - Small package

Camera / webcam
- Internal or external
  - Commonly external on desktops, internal on laptops/tablets/phones
- Accessible from multimedia applications, photo utilities,and video conferencing
  - Real-time video communication

Docking station
  - Use external keyboard and mouse
  - Extend existing laptop interfaces
  - Add additional functionality
  - Desktop adapter cards
  - Avoid cable issues

Port replicator
- Port replicator
  - Similar to a docking station
  - Does not commonly have an expansion card option
  - Usually connects using USB

Trackpad
- Replace the mouse
  - Useful in tight working areas
  - Common on laptops
- External options
  - Battery powered
  - Bluetooth connected
- Drag and tap
  - Or use multiple finger input for right-clicking, zooming, and window control
- Enable and disable
  - Avoid inadvertent mouse clicks and movements

Drawing pad
- Use an active stylus with an external digitizer
  - Very precise input

1.3 - Mobile Device Networks

Cellular networks
- Mobile devices
  - "Cell" phones
- Separate land into "cells"
  - Antenna coverages a cell with certain frequencies
- Enable or disable voice and/or data
  - Airplane mode disables all cellular networks
  - Could also disable data but keep voice

3G technology
- 3rd Generation
  - Introduced in 1998
- Upgraded data connectivity
  - Incremental 3G updates improved speeds
  - Usually several megabits per second
- Bandwidth improvement allowed new functionality
  - GPS
  - Mobile television
  - Video on demand
  - Video conferencing

4G and LTE
- Long Term Evolution (LTE)
  - A "4G" technology
  - Based on GSM and EDGE (Enhanced Data Rates for GSM Evolution)
  - Standard supports download rates of 150 Mbit/s
- LTE Advanced (LTE-A)
  - Standard supports download rates of 300 Mbit/s

5G
- Fifth generation cellular networking
  - Launched worldwide in 2020
- Significant performance improvements
  - At higher frequencies
  - Eventually 10 gigabits per second
  - Slower speeds from 100-900 Mbit/s
- Significant IoT impact
  - Bandwidth becomes less of a constraint
  - Larger data transfers
  - Faster monitoring and notification
  - Additional cloud processing

Wi-Fi
- Connect to a local 802.11 network
  - Data access for email and Internet
  - Higher speeds, no cellular bandwidth use
- Requires a local access point
  - Limited distance coverage
- May also support voice communication
  - Wi-Fi calling
  - Requires support from the voice carrier
  - Very useful when cellular coverage may not be available

Hotspot
- Turn your phone into a WiFi hotspot
  - Your own personal wireless router
  - Extend the cellular data network to all of your devices
- Dependent on phone type and provider
  - May require additional charges and data costs

Subscriber Identity Module (SIM)
- A unique identifier for cellular devices
  - Often a tiny physical card
  - Replace the SIM and the phone has a new number
- Contains information for the carrier and user
  - SIM ID and phone number
  - Cellular network information
  - Storage space for contacts, messages, etc.
- eSIM - embedded SIM
  - SIM is inside the device
  - Cannot be physically removed
- Many devices support multiple SIMs

Bluetooth pairing
- Connect a Bluetooth device
  - Built-in security - Use or verify a PIN
- Pair the devices one time
  - Future connections should be automatic
- Check with the manufacturer
  - May prefer a specific sequence
  - Discoverable mode isn't always obvious

Bluetooth pairing process
- Enable Bluetooth on both devices
  - Android and iOS: Settings / Bluetooth
- Set devices to discoverable mode
  - May require key sequence on Bluetooth device
- Select discovered device
  - Many devices may appear!
- Enter or confirm PIN
  - Should be the same on both devices
- Test connectivity
  - Devices should now communicate

GPS (Global Positioning System)
- Created by the U.S. Department of Defense
  - Over 30 satellites currently in orbit
- Precise navigation
  - Need to see at least 4 satellites
- Determines location based on timing differences
  - Longitude, latitude, altitude
- Mobile device location services and geotracking
  - Maps, directions
  - Determine physical location based on GPS, WiFi, and cellular towers

1.3 - Mobile Device Management

MDM (Mobile Device Management)
- Manage company-owned and user-owned mobile devices
  - BYOD - Bring Your Own Device
- Centralized management of the mobile devices
  - Specialized functionality
- Set policies on apps, data, camera, etc.
  - Control the remote device
  - The entire device or a "partition"
- Manage access control
  - Force screen locks and PINs on these single user devices

BYOD
- Bring Your Own Device
  - Bring Your Own Technology
- Employee owns the device
  - Need to meet the company's requirements
- Difficult to secure
  - It's both a home device and a work device
  - How is data protected?
  - What happens to the data when a device is sold or traded in?

COPE
- Corporate owned, personally enabled
  - Company buys the device
  - Used as both a corporate device and a personal device
- Organization keeps full control of the device
  - Similar to company-owned laptops and desktops
- Information is protected using corporate policies
  - Information can be deleted at any time
- CYOD - Choose Your Own Device
  - Similar to COPE, but with the user's choice of device

MDM policy enforcement
- Corporate email configuration
  - User does not need to configure anything
  - The MDM makes the changes on the device
  - Account details, server address, communication method
- Two-factor authentication
  - Require specific authentication types
  - Biometrics, pseudo-random authentication app
- Corporate applications
  - Allow or restrict app installation
  - Prevent unauthorized app usage

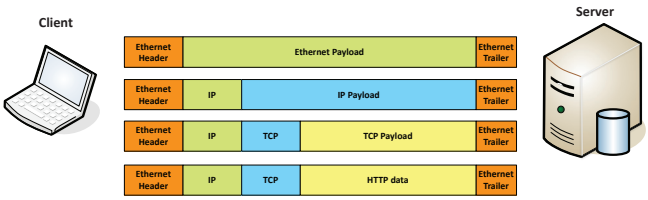
Mobile device synchronization
- Many settings are preconfigured
  - Telephone / Text messaging
- Email
  - Everyone handles email services differently
  - Corporate email configurations can vary
- Data synchronization
  - Based on data rates and speeds
  - Important for backup and recovery

Synchronizing data
- Data types
  - Calendar
  - Contacts
- Data caps and transfer costs
  - May have significant financial consequences
  - Cellular/Mobile vs 802.11
- Verify data restrictions with the wireless carrier
  - Get it in writing
- Enable or disable network connections
  - Control the use of cellular downloads

Business applications
- Mail, cloud storage, etc.
  - Many different services available
- Authenticate
  - Username, password
- Select the items to synchronize
  - Changes will appear on the mobile device
- Same process for other services
  - Often built-in to the mobile device

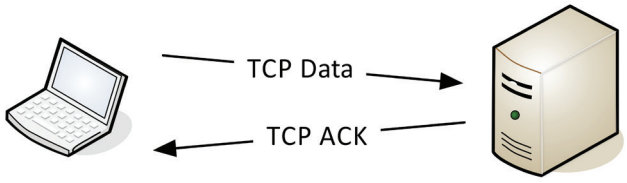
2.1 - Introduction to IP

A series of moving vans
- Efficiently move large amounts of data
  - Use a shipping truck
- The network topology is the road
  - Ethernet, DSL, cable system
- The truck is the Internet Protocol (IP)
  - We've designed the roads for this truck
- The boxes hold your data
  - Boxes of TCP and UDP
- Inside the boxes are more things
  - Application information



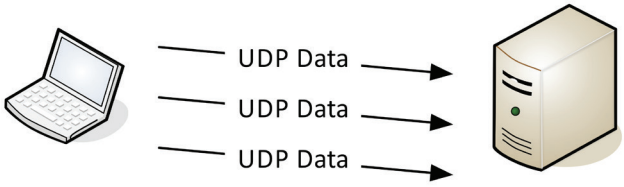
TCP and UDP
- Transported inside of IP
  - Encapsulated by the IP protocol
- Two ways to move data from place to place
  - Different features for different applications
- OSI Layer 4
  - The transport layer
- Multiplexing
  - Use many different applications at the same time
  - TCP and UDP

TCP - Transmission Control Protocol Communication



TCP – Transmission Control Protocol
- Connection-oriented
  - A formal connection setup and close
- "Reliable" delivery
  - Recovery from errors
  - Manages out-of-order messages or retransmissions
- Flow control
  - The receiver can manage how much data is sent

UDP - User Datagram Protocol Communication



UDP – User Datagram Protocol
- Connectionless - No formal open/close to the connection
- "Unreliable" delivery
  - No error recovery, no reordering of data or retransmissions
- No flow control
  - Sender determines the amount of data transmitted

Why would you ever use UDP?
- Real-time communication
  - There's no way to stop and resend the data
  - Time doesn't stop for your network
- Connectionless protocols
  - DHCP (Dynamic Host Configuration Protocol)
  - TFTP (Trivial File Transfer Protocol)
- The data might not get through
  - The application keeps track and decides what to do
  - It might not do anything

Communication using TCP
- Connection-oriented protocols prefer a "return receipt"
  - HTTPS (Hypertext Transfer Protocol Secure)
  - SSH (Secure Shell)
- The application doesn't worry about out of order frames or missing data
  - TCP handles all of the communication overhead
  - The application has one job

Speedy delivery
- The IP delivery truck delivers from one (IP) address to another (IP) address
  - Every house has an address, every computer has an IP address
- Boxes arrive at the house / IP address
  - Where do the boxes go? - Each box has a room name
- Port is written on the outside of the box
  - Drop the box into the right room

Lots of ports
- IPv4 sockets
  - Server IP address, protocol, server application port #
  - Client IP address, protocol, client port number
- Non-ephemeral ports – permanent port numbers
  - Ports 0 through 1,023 - Usually on a server or service
- Ephemeral ports – temporary port numbers
  - Ports 1,024 through 65,535
  - Determined in real-time by the client

Port numbers
- TCP and UDP ports can be any number between 0 and 65,535
- Most servers (services) use non-ephemeral (not-temporary) port numbers
  - This isn't always the case - it's just a number.
- Port numbers are for communication, not security
- Service port numbers need to be "well known"
- TCP port numbers aren't the same as UDP port numbers

2.1 - Common Ports

Port numbers
- Well-known port number
  – Client and server need to match
- Important for firewall rules
  – Port-based security
- A bit of rote memorization
  – Becomes second nature after a while
- Make sure you know port number, protocol, and how the protocol is used

FTP - File Transfer Protocol
- **tcp/20** (active mode data), **tcp/21** (control)
  – Transfers files between systems
- Authenticates with a username and password
  – Some systems use a generic/anonymous login
- Full-featured functionality - List, add, delete, etc.

SSH - Secure Shell
- Encrypted communication link - **tcp/22**
- Looks and acts the same as Telnet

Telnet
- Telnet – Telecommunication Network - **tcp/23**
- Login to devices remotely
  – Console access
- In-the-clear communication
  – Not the best choice for production systems

SMTP - Simple Mail Transfer Protocol
- SMTP - Simple Mail Transfer Protocol
  – Server to server email transfer
  – **tcp/25**
- Also used to send mail from a device to a mail server
  – Commonly configured on mobile devices and email clients
- Other protocols are used for clients to receive email
  – IMAP, POP3

DNS - Domain Name System
- Converts names to IP addresses - **udp/53**
  – www.professormesser.com = 162.159.246.164
- These are very critical resources
  – Usually multiple DNS servers are in production

DHCP - Dynamic Host Configuration Protocol
- Automated configuration of IP address, subnet mask and other options
  – **udp/67, udp/68**
  – Requires a DHCP server
  – Server, appliance, integrated into a SOHO router, etc.
- Dynamic / pooled
  – IP addresses are assigned in real-time from a pool
  – Each system is given a lease and must renew at set intervals
- DHCP reservation
  – Addresses are assigned by MAC address in the DHCP server
  – Manage addresses from one location

HTTP and HTTPS
- Hypertext Transfer Protocol
  – Communication in the browser
  – And by other applications
- In the clear or encrypted
  – Supported by nearly all web servers and clients

POP3 / IMAP
- Receive emails from an email server
  – Authenticate and transfer
- POP3 - Post office Protocol version 3
  – **tcp/110**
  – Basic mail transfer functionality
- IMAP4 - Internet Message Access Protocol v4
  – **tcp/143**
  – Includes email inbox management from multiple clients
- **SMB - Server Message Block**
  - Protocol used by Microsoft Windows
  – File sharing, printer sharing
  – Also called CIFS (Common Internet File System)
- Using NetBIOS over TCP/IP (Network Basic Input/Output System)
  – **udp/137** - NetBIOS name services (nbname)
  – **tcp/139** - NetBIOS session service (nbsession)
- Direct over **tcp/445** (NetBIOS-less)
  – Direct SMB communication over TCP without the NetBIOS transport

LDAP/LADPS
- LDAP (Lightweight Directory Access Protocol)
  – **tcp/389**
  – Store and retrieve information in a network directory
  – Commonly used in Microsoft Active Directory

RDP - Remote Desktop Protocol
- Share a desktop from a remote location over **tcp/3389**
- Remote Desktop Services on many Windows versions
- Can connect to an entire desktop or just an application
- Clients for Windows, macOS, Linux, Unix, iPhone, Android, and others

2.1 - Common Ports (continued)

Protocol	Port	Name	Description
FTP	tcp/20, tcp21	File Transfer Protocol	Sends and receives files between systems
SSH	tcp/22	Secure Shell	Encrypted console access
Telnet	tcp/23	Telecommunication Network	Insecure console access
SMTP	tcp/25	Simple Mail Transfer Protocol	Transfer email between mail servers
DNS	udp/53, tcp/53	Domain Name System	Convert domain names to IP addresses
DHCP	udp/67, udp/68	Dynamic Host Configuration Protocol	Automate the IP address configuration process
HTTP	tcp/80	Hypertext Transfer Protocol	Web server communication
HTTPS	tcp/443	Hypertext Transfer Protocol Secure	Web server communication with encryption
POP3	tcp/110	Post Office Protocol version 3	Receive email into a email client
IMAP4	tcp/143	Internet Message Access Protocol v4	A newer email client protocol
NetBIOS	udp/137	NetBIOS name service	Register, remove, and find Windows services by name
NetBIOS	tcp/139	NetBIOS session service	Windows connection-oriented data transfer
SMB	tcp/445	Server Message Block	Windows file transfers and printer sharing
LDAP	tcp/389	Lightweight Directory Access Protocol	Directory services
RDP	tcp/3389	Remote Desktop Protocol	Graphical display of remote devices

2.2 - Wireless Network Technologies

Wireless technologies

- IEEE standards
  - Institute of Electrical and Electronics Engineers
  - 802.11 committee
  - Everyone follows these standards
- Also referenced as a generation
  - 802.11ac is Wi-Fi 5
  - 802.11ax is Wi-Fi 6 and Wi-Fi 6E (extended)
  - 802.11be is Wi-Fi 7
  - Future versions will increment accordingly

802.11 technologies

- Frequencies
  - 2.4 GHz, 5 GHz, and 6 GHz - Sometimes a combination
- Channels
  - Groups of frequencies, numbered by the IEEE
  - Frequencies are managed by governing entities
  - Using non-overlapping channels would be optimal
- Bandwidth
  - Amount of frequency in use
  - 20 MHz, 40 MHz, 80 MHz, 160 MHz, etc.

Bluetooth

- Remove the wires
  - Headsets, speakers, keyboards / mice
- Uses the 2.4 GHz range
  - Unlicensed ISM (Industrial, Scientific and Medical) band
  - Same as 802.11
- Short-range
  - Most consumer devices operate to about 10 meters

RFID (Radio-frequency identification)

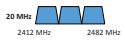
- It's everywhere
  - Access badges
  - Inventory/Assembly line tracking
  - Pet/Animal identification
  - Anything that needs to be tracked
- Radar technology
  - Radio energy transmitted to the tag
  - RF powers the tag, ID is transmitted back
  - Bidirectional communication
  - Some tag formats can be active/powered

NFC (Near field communication)

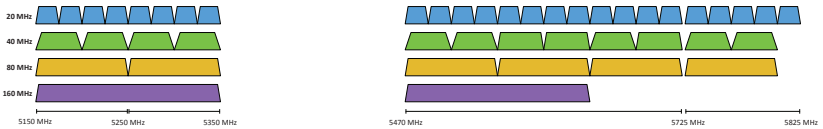
- Two-way wireless communication
  - Builds on RFID, which is mostly one-way
- Payment systems - Major credit cards, online wallets
- Bootstrap for other wireless
  - NFC helps with Bluetooth pairing
- Access token, identity "card"
  - Short range with encryption support

2.2 - Wireless Network Technologies (continued)

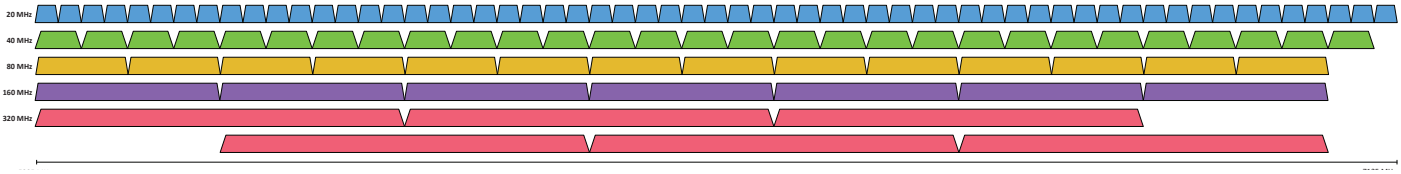
2.4 GHz Spectrum for 802.11 - North America



5 GHz Spectrum for 802.11 - North America



6 GHz Spectrum for 802.11 - North America



2.3 - Network Services

DNS server
- Domain Name System
  - Convert names to IP addresses
  - And vice versa
- Distributed naming system
  - The load is balanced across many different servers
- Usually managed by the ISP or IT department
  - A critical resource

DHCP server
- Dynamic Host Configuration Protocol
  - Automatic IP address configuration
- Very common service
  - Available on most home routers
- Enterprise DHCP will be redundant
  - Usually running on central servers

File share
- Centralized storage of documents,spreadsheets, videos, pictures, and any other files
  - A fileshare
- Standard system of file management
  - SMB (Server Message Block), Apple Filing Protocol (AFP), etc.
- The front-end hides the protocol
  - Copy, delete, rename, etc.

Print server
- Connect a printer to the network
  - Provide printing services for all network devices
- May be software in a computer
  - Computer is connected to the printer
- May be built-in to the printer
  - Network adapter and software
- Uses standard printing protocols
  - SMB (Server Message Block), IPP (Internet Printing Protocol), LPD (Line Printer Daemon)

Mail server
- Store your incoming mail
  - Send your outgoing mail
- Usually managed by the ISP or the IT department
  - A complex set of requirements
- Usually one of the most important services
  - 24 x 7 support

Syslog
- Standard for message logging
  - Diverse systems, consolidated log
- Usually a central logging receiver
  - Integrated into the SIEM
- You're going to need a lot of disk space
  - No, more. More than that.

Web server
- Respond to browser requests
  - Using standard web browsing protocols - HTTP/HTTPS
  - Pages are built with HTML, HTML5
- Web pages are stored on the server
  - Downloaded to the browser
  - Static pages or built dynamically in real-time

Authentication server
- Login authentication to resources
  - Centralized management
- Almost always an enterprise service
  - Not required on a home network
- Usually a set of redundant servers
  - Always available
  - Extremely important service

Database servers
- Database table storage
  - Save information in columns and rows
  - Similar to a spreadsheet
- Relational database
  - Link tables together
  - These links are relationships
  - Flexible and fast
- Structured Query Language (SQL)
  - Access data using a standard language
  - Microsoft SQL Server, MySQL, etc.

NTP servers
- The time of day is more important than you might think
  - Encryption, logins, backups, log timestamps
  - Everything needs to work from the same clock
- NTP server
  - Listens on udp/123
  - Responds to time requests from NTP clients
- NTP client
  - Requests time updates from NTP server
  - Daily synchronization is common

Spam gateways
- Unsolicited messages
  - Email, forums, etc.
- Various content
  - Commercial advertising
  - Non-commercial proselytizing
  - Phishing attempts
- Significant technology issue
  - Security concerns, resource utilization, storage costs, managing the spam
- Unsolicited email
  - Stop it at the gateway before it reaches the user
  - On-site or cloud-based

All-in-one security appliance
- Next-generation firewall, Unified Threat Management (UTM) / Web security gateway
- URL filter / Content inspection
- Malware inspection
- Spam filter
- CSU/DSU
- Router, Switch
- Firewall
- IDS/IPS
- Bandwidth shaper
- VPN endpoint

Load balancers
- Distribute the load
  - Multiple servers - Invisible to the end-user
- Large-scale implementations
  - Web server farms, database farms
- Fault tolerance
  - Server outages have no effect - Very fast convergence

Proxy server
- An intermediate server
  - Client makes the request to the proxy
  - The proxy performs the actual request
  - The proxy provides results back to the client
- Useful features
  - Access control, caching, URL filtering, content scanning

SCADA / ICS
- Supervisory Control and Data Acquisition System
  - Large-scale, multi-site Industrial Control Systems (ICS)
- PC manages equipment
  - Power generation, refining, manufacturing equipment
  - Facilities, industrial, energy, logistics
- Distributed control systems
  - Real-time information
  - System control
- Requires extensive segmentation
  - No access from the outside

Legacy and embedded systems
- Legacy systems
  - Another expression for “really old”
  - May also be “really important”
  - Learning old things can be just as important as learning the new things
- Embedded systems
  - Purpose-built device
  - Not usual to have direct access to the operating system
  - Alarm system, door security, time card system

IoT (Internet of Things) devices
- Appliances
  - Refrigerators
- Smart devices
  - Smart speakers respond to voice
- Air control
  - Thermostats, temperature control
- Access
  - Smart doorbells
- May require a segmented network
  - Limit any security breaches

2.4 - DNS Configuration

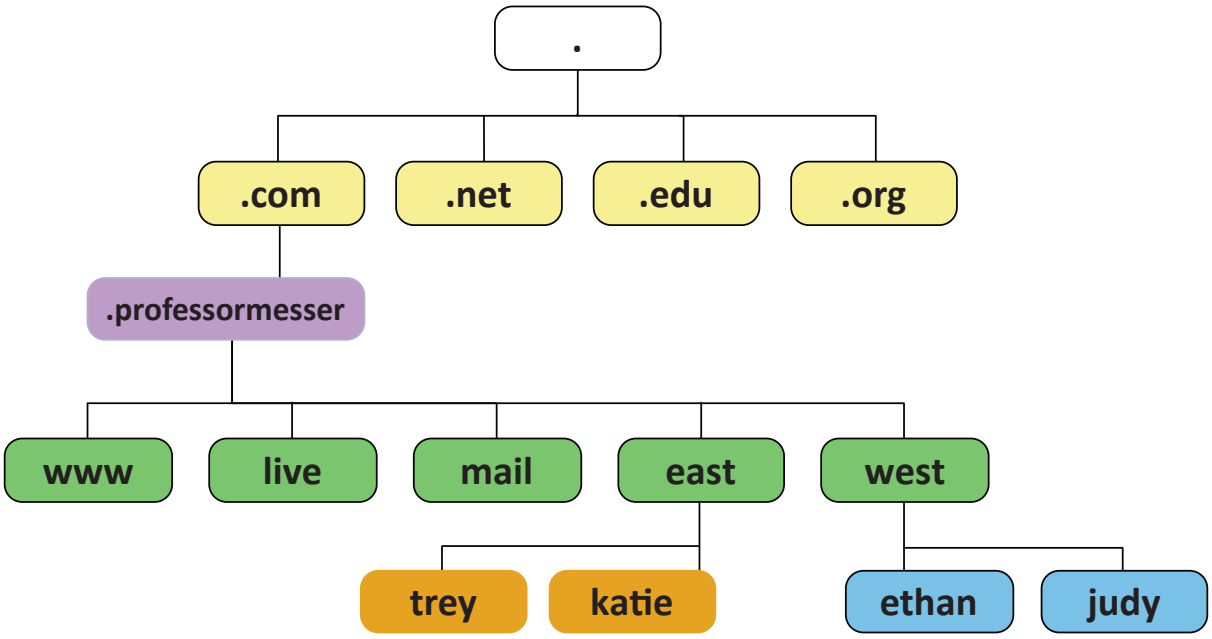
Domain Name System

- Translates human-readable names into computer-readable IP addresses
  - You only need to remember www.ProfessorMesser.com
- Hierarchical
  - Follow the path
- Distributed database
  - Many DNS servers
  - 13 root server clusters (over 1,000 actual servers)
  - Hundreds of generic top-level domains (gTLDs) - .com, .org, .net, etc.
  - Over 275 country code top-level domains (ccTLDs) - .us, .ca, .uk, etc.

DNS lookup

```
professor@Odyssey ~ % dig www.professormesser.com

; <<>> DiG 9.10.6 <<>> www.professormesser.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 58050
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.professormesser.com. IN       A

;; ANSWER SECTION:
www.professormesser.com. 300      IN       A        104.22.72.108
www.professormesser.com. 300      IN       A        172.67.41.114
www.professormesser.com. 300      IN       A        104.22.73.108

;; Query time: 42 msec
;; SERVER: 9.9.9.9#53(9.9.9.9)
;; WHEN: Thu Jun 17 09:17:26 EDT 2021
;; MSG SIZE  rcvd: 100
```

The DNS hierarchy



DNS records

- Resource Records (RR)
  - The database records of domain name services
- Over 30 record types
  - IP addresses, certificates, host alias names, etc.
- These are important and critical configurations
  - Make sure to check your settings, backup, and test!

Address records (A) (AAAA)

- Defines the IP address of a host
  - This is the most popular query
- A records are for IPv4 addresses
  - Modify the A record to change the host name to IP address resolution
- AAAA records are for IPv6 addresses
  - The same DNS server, different records

```
www.professormesser.com.    IN A    162.159.246.164  ; Professor Messer
```

Canonical name records (CNAME)

- A name is an alias of another, canonical name
  - One physical server, multiple services

```
; Alias (canonical) names
    chat     IN CNAME     mail.example.com.
    ftp      IN CNAME     mail.example.com.
    www      IN CNAME     mail.example.com.
```

2.4 - DNS Configuration (continued)

Mail exchanger record (MX)

- Determines the host name for the mail server - this isn't an IP address; it's a name

```
; This is the mail-exchanger.  You can list more than one (if
; applicable), with the integer field indicating priority (lowest
; being a higher priority)
      IN MX       mail.mydomain.name.

; A list of machine names & addresses
    jack.mydomain.name.     IN A    123.12.41.40    ; Windows 10
    mail.mydomain.name.     IN A    123.12.41.41    ; Linux (main server)
    sam.mydomain.name.      IN A    123.12.41.42    ; Windows 11
```

Text records (TXT)

- Human-readable text information
  - Useful public information
  - Was originally designed for informal information
- Can be used for verification purposes
  - If you have access to the DNS, then you must be the administrator of the domain name
- Commonly used for email security
  - External email servers validate information from your DNS

Viewing TXT records with nslookup

```
professor@Odyssey ~ % nslookup -type=txt google.com
;; Truncated, retrying in TCP mode.
Server:         9.9.9.9
Address:  9.9.9.9#53

Non-authoritative answer:
google.com          text = "docusign=1b0a6754-49b1-4db5-8540-d2c12664b289"
google.com          text = "v=spf1 include:_spf.google.com ~all"
google.com          text = "apple-domain-verification=30afIBcvSuDV2PLX"
google.com          text = "globalsign-smime-dv=CDYX+XFHUw2wml6/Gb8+59BsH31KzUr6c1l2BPvqKX8="
google.com          text = "facebook-domain-verification=22rm551cu4k0ab0bxsw536tlds4h95"
google.com          text = "google-site-verification=TV9-DBe4R80X4v0M4U_bd_
J9cpOJM0nikft0jAgjmsQ"
google.com          text = "MS=E4A68B9AB2BB9670BCE15412F62916164C0B20BB"
google.com          text = "docusign=05958488-4752-4ef2-95eb-aa7ba8a3bd0e"
google.com          text = "google-site-verification=wD8N7i1JTNTkezJ49swvWW48f8_9xveREV4oB-
0Hf5o"
```

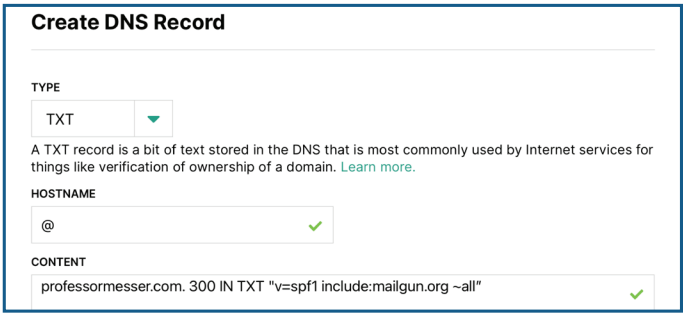
Domain Keys Identified Mail (DKIM)

- Digitally sign a domain's outgoing mail
  - Validated by mail servers, not usually seen by the end user
  - The public key is in the DKIM TXT record

```
; DKIM TXT records
; owner  class  ttl  TXT "attribute-name=attribute value"
1517680427.professormesser._domainkey.professormesser.com. IN 300 TXT
("v=DKIM1;t=s;p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDqCUQ5dpKOtwQdE2k8HaCQqV+f"
        "3y3OBCzNz75IfEXtk+sTBiDcGWIcapUzkgC4tNOboHBw57APzNInmjH9yZn15TB"
        "TfTavC44nXidUZ8LzsJGWVvYYxoFR5DuBoi/zIOOHv6YDUpDxJa9knZABTOWLS2F"
        "YtK9dWAMaOZdtTBOhQIDAQAB")
```

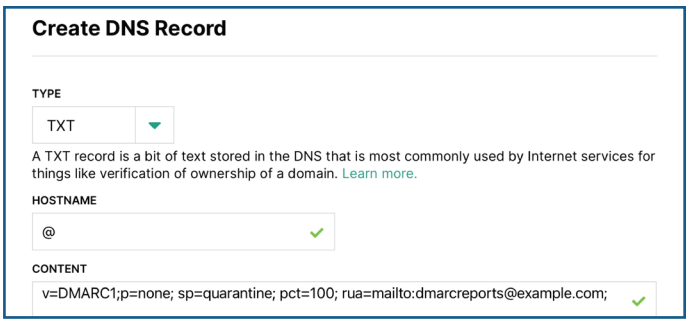
Sender Policy Framework (SPF)

- SPF protocol
  - A list of all servers authorized to send emails for this domain
  - Prevent mail spoofing
  - Mail servers perform a check to see if incoming mail really did come from an authorized host



DMARC

- Domain-based Message Authentication, Reporting, and Conformance (DMARC)
  - Prevent unauthorized email use (spoofing)
  - An extension of SPF and DKIM
- You decide what external email servers should do with emails that don't validate through SPF or DKIM
  - That policy is written into a DMARC TXT record
  - Accept all, send to spam, or reject the email
  - Compliance reports can be sent to the email administrator



2.4 - DHCP

DCHP

- IPv4 address configuration used to be manual
  - IP address, subnet mask, gateway, DNS servers, NTP servers, etc.
- Needed a more automatic method
  - Scale to thousands of systems
  - Hands-off IP configurations
- Dynamic Host Configuration Protocol (DHCP)
  - Initially released in 1997, updated through the years
  - Provides automatic address / IP configuration
  - Used for almost all devices

DHCP leases

- DORA
  - A four-step process
- Discover
  - Find a DHCP server
- Offer
  - Get an offer
- Request
  - Lock in the offer
- Acknowledge
  - DHCP server confirmation

DHCP scopes

- Configured on the DHCP server
- IP address range
  - And excluded addresses
- Subnet mask
- Lease durations
- Other scope options
  - DNS server
  - Default gateway
  - VOIP servers

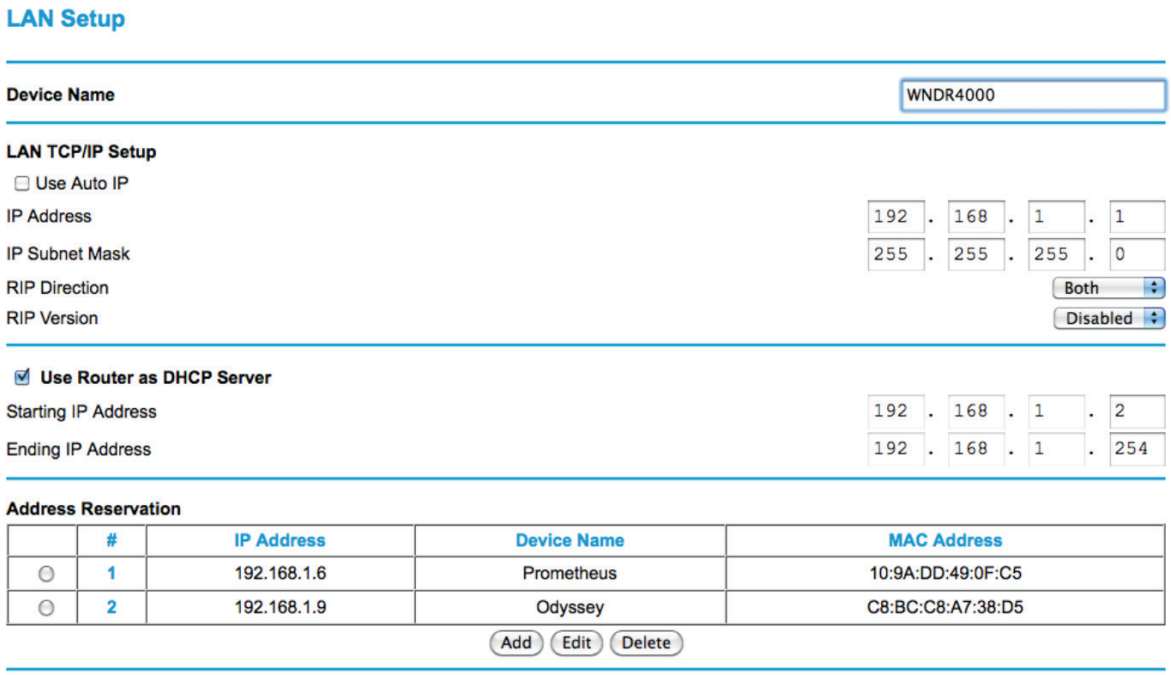
DHCP pools

- Grouping of IP addresses
  - Each subnet has its own scope
  - 192.168.1.0/24
  - 192.168.2.0/24
  - 192.168.3.0/24
  - ...
- A scope is generally a single contiguous pool of IP addresses
  - DHCP exceptions can be made inside of the scope

Address reservation

- Address reservation
  - Administratively configured
- Table of MAC addresses
  - Each MAC address has a matching IP address
- Other names
  - Static DHCP Assignment
  - Static DHCP
  - IP Reservation

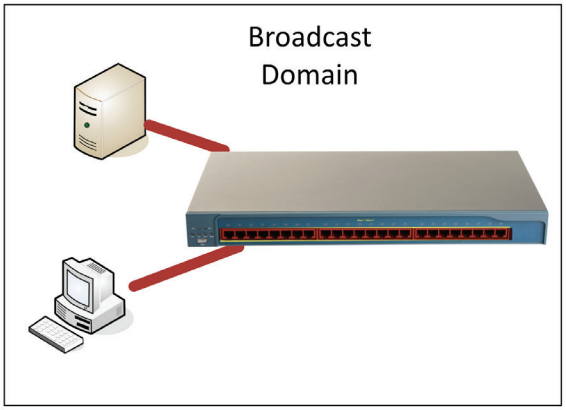
Address Reservation

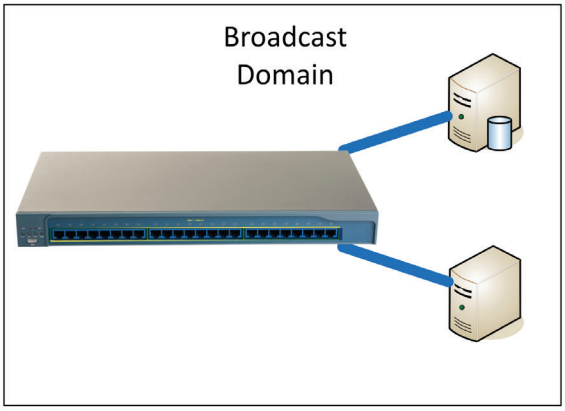


2.4 - VLANs and VPNs

LANs

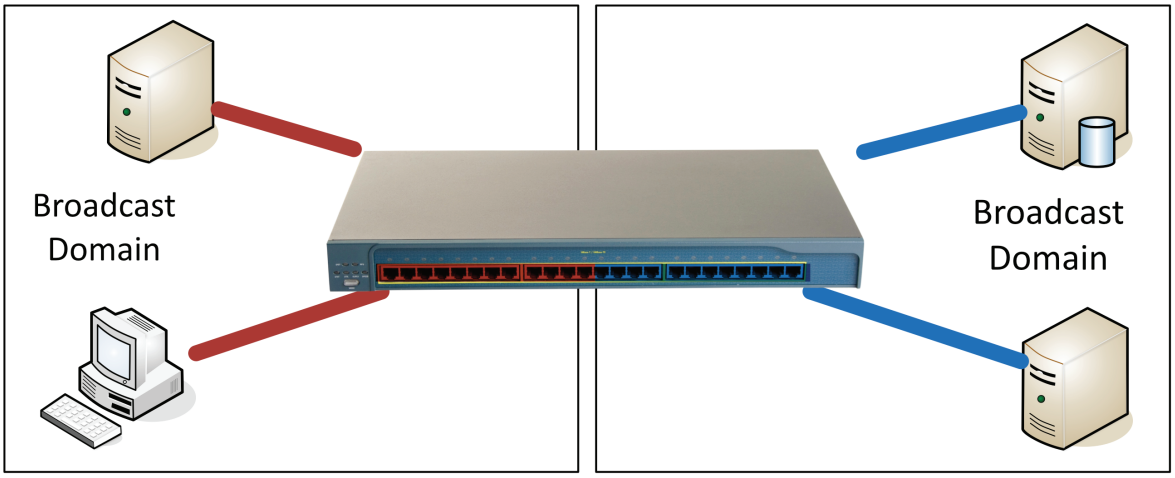
- Local Area Networks
  - A group of devices in the same broadcast domain





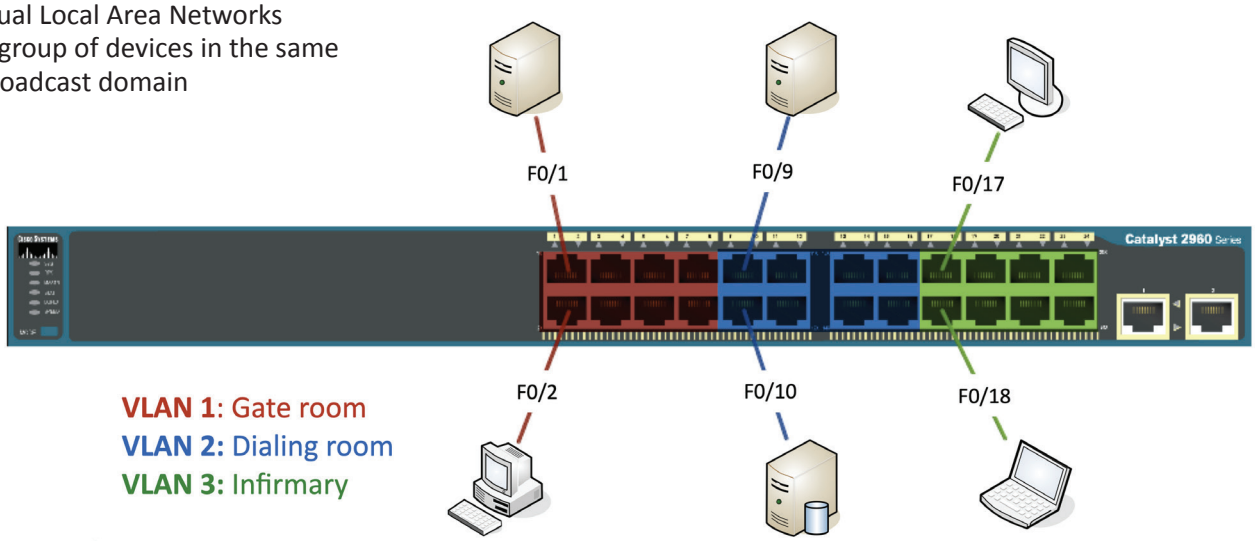
Virtual LANs

- Virtual Local Area Networks
  - A group of devices in the same broadcast domain
  - Separated logically instead of physically



Configuring VLANs

- Virtual Local Area Networks
  - A group of devices in the same broadcast domain



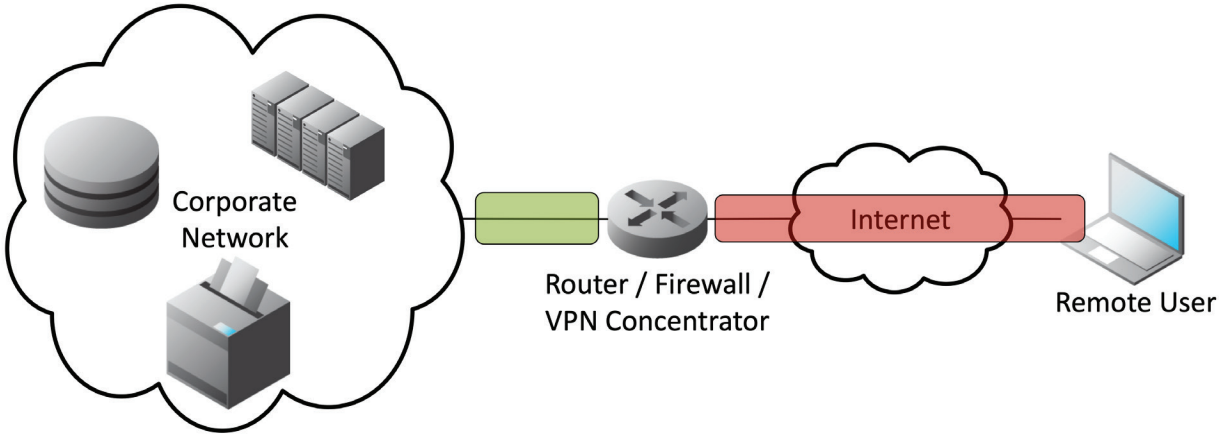
2.4 - VLANs and VPNs (continued)

VPNs
- Virtual Private Networks
  - Encrypted (private) data traversing a public network
- Concentrator
  - Encryption/decryption access device
  - Often integrated into a firewall
- Many deployment options
  - Specialized cryptographic hardware
  - Software-based options available
- Used with client software
  - Sometimes built into the OS

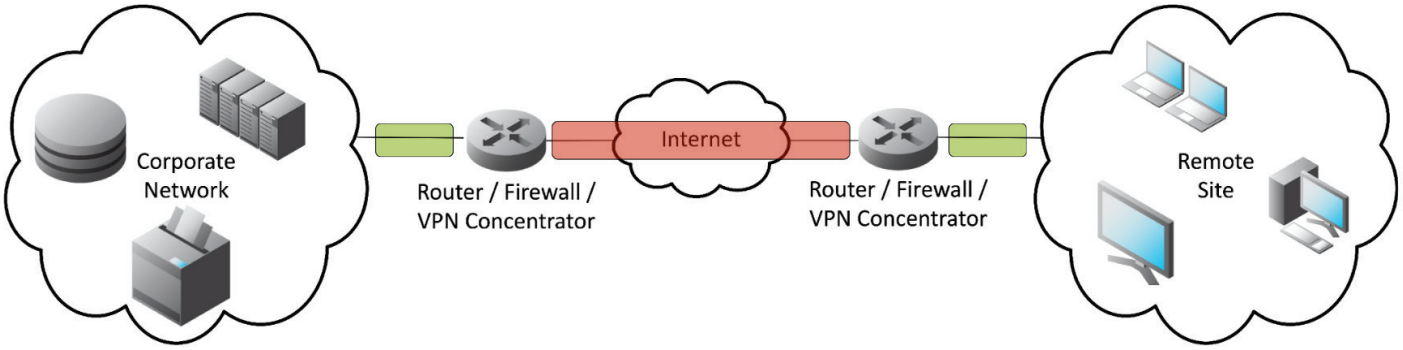
Client-to-Site VPN
- On-demand access from a remote device
  - Software connects to a VPN concentrator
- Some software can be configured as always-on

Site-to-site VPN
- Always-on
  - Or almost always
- Firewalls often act as VPN concentrators
  - Probably already have firewalls in place

Client-to-Site VPN



Site-to-site VPN



2.5 - Network Devices

Network devices
- Many different devices and components
  - All have different roles
- Some of these functions are combined together
  - Wireless router/switch/firewall
- Compare different devices
  - Understand when they should be used

Routers
- Routes traffic between IP subnets
  - Makes forwarding decisions based on IP address
  - Routers inside of switches sometimes called "layer 3 switches"
- Often connects diverse network types
  - LAN, WAN, copper, fiber

Switches
- Bridging done in hardware
  - Application-specific integrated circuit (ASIC)
  - Forwards traffic based on data link address
- Many ports and features
  - The core of an enterprise network
  - May provide Power over Ethernet (PoE)
- Multilayer switch
  - Includes routing functionality

2.5 - Network Devices (continued)

Unmanaged switches
- Very few configuration options - plug and play
- Fixed configuration - no VLANs
- Very little integration with other devices
  - No management protocols
- Low price point - simple is less expensive

Managed switches
- VLAN support
  - Interconnect with other switches via 802.1Q
- Traffic prioritization
  - Voice traffic gets a higher priority
- Redundancy support
  - Spanning Tree Protocol (STP)
- Port mirroring
  - Capture packets
- External management
  - Simple Network Management Protocol (SNMP)

Access point
- Not a wireless router
  - A wireless router is a router and an access point in a single device
- An access point is a bridge
  - Extends the wired network onto the wireless network
  - Makes forwarding decisions based on MAC address

Patch Panels
- Combination of punch-down blocks and RJ-45 connectors
- Runs from desks are made once
  - Permanently punched down to patch panel
- Patch panel to switch can be easily changed
  - No special tools - use existing cables

Firewalls
- Filters traffic by port number
  - OSI layer 4 (TCP/UDP)
  - Some firewalls can filter based on the application
- Can encrypt traffic into/out of the network
  - Protect your traffic between sites
- Can proxy traffic
  - A common security technique
- Most firewalls can be layer 3 devices (routers)
  - Usually sits on the ingress/egress of the network

Power over Ethernet (PoE)
- Power provided on an Ethernet cable
  - One wire for both network and electricity
  - Phones, cameras, wireless access points
  - Useful in difficult-to-power areas
- Power provided at the switch
  - Built-in power - Endspans
  - In-line power injector - Midspans

PoE switch
- Power over Ethernet
  - Commonly marked on the switch or interfaces

PoE, PoE+, PoE++
- PoE
  - The original PoE specification
  - Now part of the 802.3 standard
  - 15.4 watts DC power, 350 mA max current
- PoE+
  - Now also part of the 802.3 standard
  - 25.5 watts DC power, 600 mA max current
- PoE++
  - 51 W (Type 3), 600 mA max current
  - 71.3 W (Type 4), 960 mA max current
  - PoE with 10GBASE-T
- Compare the device with the switch support
  - PoE+ won't power a PoE++ device

Cable modem
- Broadband
  - Transmission across multiple frequencies
  - Different traffic types
- Data on the "cable" network
  - DOCSIS (Data Over Cable Service Interface Specification)
- High-speed networking
  - Speeds up to 1 Gigabit/s are available
- Multiple services - Data, voice, video

DSL modem
- Digital Subscriber Line - uses telephone lines
- Download speed is faster than the upload speed (asymmetric)
  - ~10,000 foot limitation from the central office (CO)
  - Faster speeds may be possible if closer to the CO

ONT
- Optical network terminal - fiber to the premises
- Connect the ISP fiber network to the copper network
  - Demarcation point (demarc) in the data center
  - Terminal box on the side of the building
- Line of responsibility
  - One side of the box is the ISP
  - Other side of the box is your network

Network Interface Card (NIC)
- The fundamental network device
  - Every device on the network has a NIC
  - Computers, servers, printers, routers, switches, phones, tablets, cameras, etc.
- Specific to the network type - Ethernet, WAN, wireless
- Often built-in to the motherboard
  - Or added as an expansion card
- Contains the hardware address
  - Media Access Control (MAC) address
  - A unique hardware designation

2.6 - IPv4 and IPv6

IP addressing

- IPv4 is the primary protocol for everything we do
  - You probably won't configure anything else
- IPv6 is now part of all major operating systems
  - And the backbone of our Internet infrastructure

IPv4 addresses

- Internet Protocol version 4
  - OSI Layer 3 address

Public IPv4 addresses

- Each IPv4 address on the Internet is unique
  - 1.1.1.1 can communicate to 2.2.2.2
- It is estimated that there are over 20 billion devices connected to the Internet (and growing)
  - IPv4 supports around 4.29 billion addresses
  - There's an obvious scalability issue
- We've found ways to manage the demand
  - Network Address Translation (NAT)

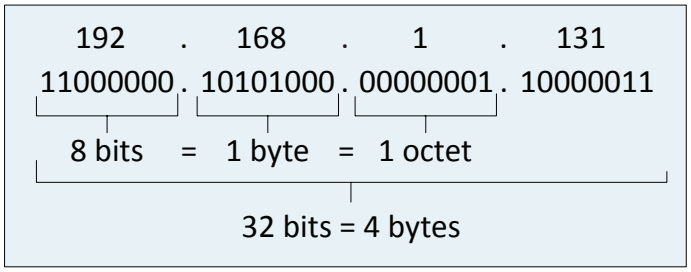
Private IP address ranges

- Large private IP address ranges
  - Properly design and scale large networks
- Private IP addresses are not Internet-routable
  - But can be routed internally
  - Use NAT for everything else
- Defined in RFC 1918
  - Request for Comment

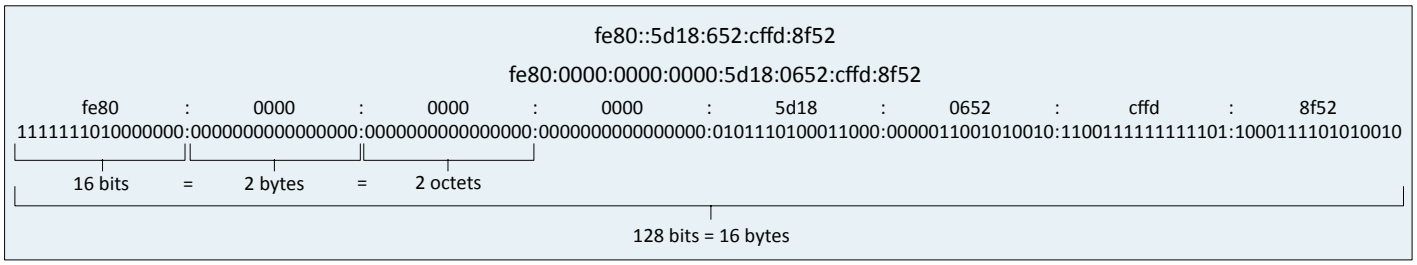
IPv6 addresses

- Internet Protocol v6 - 128-bit address
  - 340,282,366,920,938,463,463,374,607, 431,768,211,456 addresses (340 undecillion)
  - 6.8 billion people could each have 5,000,000,000,000,000,000,000,000,000 addresses
- Your DNS is very important!
- First 64 bits is generally the network prefix (/64)
- Last 64 bits is then the host network address

IPv4 addresses format



IPv6 addresses format



RFC 1918 private IP addresses

IP address range	Number of addresses	Classful description	Largest CIDR block (subnet mask)	Host ID size
10.0.0.0 – 10.255.255.255	16,777,216	single class A	10.0.0.0/8 (255.0.0.0)	24 bits
172.16.0.0 – 172.31.255.255	1,048,576	16 contiguous class Bs	172.16.0.0/12 (255.240.0.0)	20 bits
192.168.0.0 – 192.168.255.255	65,536	256 contiguous class Cs	192.168.0.0/16 (255.255.0.0)	16 bits

2.6 - Assigning IP Addresses

Networking with IPv4

- IP Address, e.g., 192.168.1.165
  - Every device needs a unique IP address
- Subnet mask, e.g., 255.255.255.0
  - Used by the local device to determine what subnet it's on
  - The subnet mask isn't (usually) transmitted across the network
  - You'll ask for the subnet mask all the time
  - What's the subnet mask of this network?
- Default gateway, e.g., 192.168.1.1
  - The router that allows you to communicate outside of your local subnet
  - The default gateway must be an IP address on the local subnet

Static IP addressing

- An IP address that doesn't change
  - Without a manual configuration change
- Manually configure the IP address
  - Type it from the keyboard on each device
- Can be difficult to manage
  - If the address changes, you must visit the device
  - This obviously does not scale
- Try to avoid using manual IP addressing
  - There are better ways to assign addresses
  - DHCP reservations

Turning dynamic into static

- DHCP assigns an IP address from the first available from a large pool of addresses
  - Your IP address will occasionally change
- You may not want your IP address to change
  - Server, printer, or personal preference
- Disable DHCP on the device
  - Configure the IP address information manually
  - Requires additional administration
- Better: Configure an IP reservation on the DHCP server
  - Associate a specific MAC address with an IP address

Automatic Private IP Addressing (APIPA)

- A link-local address - No forwarding by routers
- APIPA uses 169.254.0.0 through 169.254.255.255
  - First and last 256 addresses are reserved
  - Functional block of 169.254.1.0 through 169.254.254.255
- Randomly assigned in this block
  - Uses ARP to confirm the address isn't currently in use

2.7 - Internet Connection Types

Satellite networking

- Communication to a satellite
  - Non-terrestrial communication
- High cost relative to terrestrial networking
  - 100 Mbit/s down, 5 Mbit/s up are common
  - Remote sites, difficult-to-network sites
- High latency
  - Some systems have 250 ms up, 250 ms down
  - Starlink advertises 25 ms to 60 ms
- Not always a perfect option
  - Line of sight, rain fade

Fiber

- High speed data communication
  - Frequencies of light
- Higher installation cost than copper
  - Equipment is more costly
  - More difficult to repair
  - Communicate over long distances
- Large installation in the WAN core
  - Supports very high data rates
  - SONET, wavelength division multiplexing
- Fiber is slowly approaching the premises
  - Business and home use

Cable

- Broadband
  - Transmission across multiple frequencies
  - Different traffic types
- Data on the "cable" network
  - DOCSIS (Data Over Cable Service Interface Specification)
- High-speed networking
  - 50 Mbits/s through 1,000+ Mbit/s are common
- Multiple services
  - Data, voice, video

DSL

- DSL (Digital Subscriber Line)
  - Uses telephone lines
- Download speed is faster than the upload speed (asymmetric)
  - 200 Mbit/s downstream / 20 Mbit/s upstream are common
  - ~10,000 foot limitation from the central office (CO)
  - Faster speeds may be possible if closer to the CO

2.7 - Internet Connection Types (continued)

Cellular networks
- Mobile devices
  - "Cell" phones
- Separate land into "cells"
  - Antenna covers a cell with certain frequencies
- Tethering
  - Turn your phone into a wireless router
- Mobile hotspot
  - Standalone devices
  - Use your phone for other things

WISP
- Wireless Internet Service Provider
  - Terrestrial Internet access using wireless
- Connect rural or remote locations
  - Internet access for everyone
- Many different deployment technologies
  - Meshed 802.11
  - 5G home Internet
  - Proprietary wireless
- Need an outdoor antenna
  - Speeds can range from ~ 10 to 1,000 megabits per second

2.7 - Network Types

LAN
- Local Area Network
  - Local is relative
- A building or group of buildings
  - High-speed connectivity
- Ethernet and 802.11 wireless
  - Any slower and it isn't "local"

WAN
- Wide Area Network
  - Spanning the globe
- Generally connects LANs across a distance
  - And generally much slower than the LAN
- Many different WAN technologies
  - Point-to-point serial, MPLS, etc.
  - Terrestrial and non-terrestrial

PAN
- Personal Area Network
  - Your own private network
  - Bluetooth, IR, NFC
- Automobile
  - Audio output
  - Integrate with phone
- Mobile phone
  - Wireless headset
- Health
  - Workout telemetry, daily reports

MAN
- Metropolitan Area Network
  - A network in your city
  - Larger than a LAN, often smaller than a WAN
- Historically MAN-specific topologies
  - Metro Ethernet
- Common to see government ownership
  - They "own" the right-of-way

SAN
- Storage Area Network (SAN)
  - Looks and feels like a local storage device
  - Block-level access
  - Very efficient reading and writing
- Requires a lot of bandwidth
  - May use an isolated network and high-speed network technologies

WLAN
- Wireless LAN
  - 802.11 technologies
- Mobility
  - Within a building
  - In a limited geographical area
- Expand coverage with additional access points
  - Downtown area
  - Large campus

2.8 - Network Tools

Cable crimper

- "Pinch" the connector onto a wire
  - Coaxial, twisted pair, fiber
- Connect the modular connector to the Ethernet cable
  - The final step of the process
- Metal prongs are pushed through the insulation
  - The plug is also permanently pressed onto the cable sheath

Crimping best practices

- Get a good crimper
  - And a good pair of electrician's scissors / cable snips
  - And a good wire stripper
- Make sure you use the correct modular connectors
  - Differences between wire types
- Practice, practice, practice
  - It won't take long to become proficient

WiFi analyzer

- Wireless networks are incredibly easy to monitor
  - Everyone "hears" everything
- Purpose-built hardware or mobile device add-on
  - Specializes in 802.11 analysis
- Identify errors and interference
  - Validate antenna location and installation

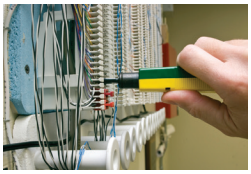
Tone generator

- Where does that wire go?
  - Follow the tone
- Tone generator
  - Puts an analog sound on the wire
- Inductive probe
  - Doesn't need to touch the copper
  - Hear through a small speaker

Using the tone generator and probe

- Easy wire tracing
  - Even in complex environments
- Connect the tone generator to the wire
  - Modular jack, coax, punch down connectors
- Use the probe to locate the sound
  - The two-tone sound is easy to find

Punch-down tool



- "Punch" a wire into a wiring block
  - 66 block, 110 block, and others
- Can be tedious
  - Every wire must be individually punched
- Trims the wires during the punch
  - Very efficient process

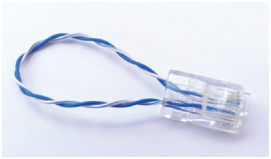
Punch-down best-practices

- Organization is key
  - Lots of wires,
  - Cable management
- Maintain your twists
  - Your Category 6A cable will thank you later
- Document everything
  - Written documentation, tags, graffiti

Cable testers

- Relatively simple
  - Continuity test
- Can identify missing pins
  - Or crossed wires
- Not usually used for frequency testing
  - Crosstalk, signal loss, etc.

Loopback plugs



- Useful for testing physical ports
  - Or fooling your applications
- Serial / RS-232 (9 pin or 25 pin)
- Network connections
  - Ethernet, T1, Fiber
- These are not cross-over cables

Taps and Port Mirrors

- Intercept network traffic
  - Send a copy to a packet capture device
- Physical taps
  - Disconnect the link, put a tap in the middle
  - Can be an active or passive tap
- Port mirror
  - Port redirection, SPAN (Switched Port ANalyzer)
  - Software-based tap
  - Limited functionality, but can work well in a pinch

3.1 - Display Types

LCD

- Liquid crystal display
  - Light shines through liquid crystals
- Advantages
  - Lightweight
  - Relatively low power
  - Relatively inexpensive
- Disadvantages
  - Black levels are a challenge
  - Requires separate backlight
  - Florescent, LED, etc.
  - Lights are difficult to replace

LCD technologies

- TN (Twisted Nematic) LCD
  - The original LCD technology
  - Fast response times (gaming!)
  - Poor viewing angles - color shifts
- IPS (In Plane Switching) LCD
  - Excellent color representation
  - Can be more expensive to produce than TN
- VA (Vertical Alignment) LCD
  - A good compromise between TN and IPS
  - Good color representation
  - Often slower response times than TN

OLED

- Organic Light Emitting Diode
  - Organic compound emits light when receiving an electric current
- No backlight - the organic compound provides the light
- Thinner and lighter
  - Flexible and mobile - no glass needed
- Tablets, phones, smart watches
  - Very accurate color representation
  - Tends to be a bit higher cost than LCD

Mini LED

- Same backlight technology as a conventional LED-backlit LCD
  - But much smaller Light Emitting Diodes (LEDs)
- Each LED can be enabled or disabled
  - And color and intensity can be different
- Much better control over dark screen areas
  - Deeper blacks, better color representation

Touchscreen

- Merge laptop and tablet input
  - Digitizer responds to touch
  - No keyboard required
  - But often still available
- Many options for input
  - Use the best one for the job

Digitizer

- Use a pen-like device as input
  - Stylus input
  - Useful for graphical input
- Used commonly on laptop / tablets
  - Or hybrid devices

Backlight and inverter

- LCD displays need a backlight
  - Florescent lamp or LED lights
- Some laptops have inverters
  - Turn DC into AC
- Verify backlight
  - Look closely
  - Use a flashlight
- May need to replace the LCD inverter or display
  - Choose carefully

3.1 - Display Attributes

Display attributes

- Not all displays are created equal
  - Various specifications, options, and settings
- Requirements can vary
  - Web browsing
  - Video and photo editing
  - Information displays
  - Phones
- Evaluate all of the technical details
  - Choose the best display for the requirement

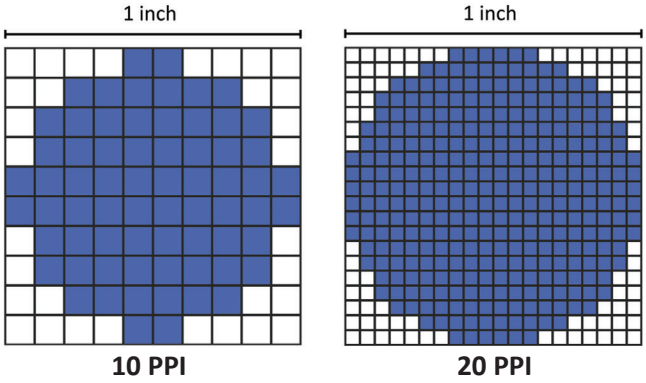
Pixel density

- Measure how many pixels in an inch
  - Or pixels per centimeter
  - Apply a physical measurement to a display
- A higher pixel density looks better
  - Higher clarity and sharpness
- Consider how the image will be viewed
  - Displays, printers (dots per inch)

3.1 - Display Attributes (continued)

Pixel density comparison

- Number of pixels / Number of inches
  - Pixels per (divided by) inch
- 27 inch (diagonal) 4K display
  - 3,840 horizontal pixels / 24 inches wide (not diagonal)
  - 160 pixels per inch
- 65 inch (diagonal) 4K television
  - 3,840 horizontal pixels / 57 inches wide
  - 67 pixels per inch

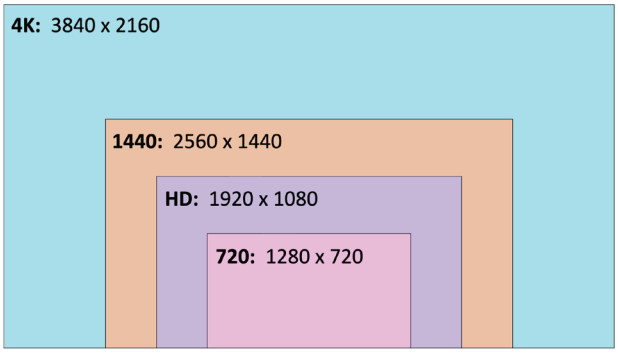


Refresh rates

- We are viewing many images shown consecutively
  - A still "frame" or image
- Usually measured in hertz (Hz)
  - Number of cycles per second - frames per second (FPS)
- Frame rates may vary
  - Movies: 24 fps
  - Television: 30 fps
  - Video games: 60 fps and higher
- Higher refresh rates are useful for rapidly moving content
  - Video games, sporting events, video production
- Maximum refresh rate is determined by the video adapter/processor and the display device
  - The connection between the two is also important
  - HDMI 2.1 supports 4K at 144 Hz
  - DisplayPort 2.1 supports dual 4K at 144 Hz

Screen resolution

- The number of pixels - width x height
- More pixels means more detailed display output
  - Perceived sharpness, a clearer view
- Many standards
  - Resolutions can vary widely - 16:9 aspect ratio is common



Color gamut

- Color gamut
  - The range of colors available on a display or output device
  - This is less than the capabilities of the human eye
- An important consideration in a display
  - Color representation
  - Range of colors
- Many standards
  - sRGB - Standard Red Green Blue
  - Adobe RGB
  - International Telecommunication Union (ITU) standards for HDTV

Color coverage

- A display may be compared with an existing color gamut standard
  - Part of the display specifications
- Gamut compatibility can vary widely
  - Higher percentages tend to be more expensive
- OLED displays provide a wide color gamut
  - Best-in-class color gamut support

3.2 - Network Cables

The importance of cable

- Fundamental to network communication
  - Incredibly important foundation
- Usually only get one good opportunity at building your cabling infrastructure - Make it good!
- The vast majority of wireless communication uses cables
  - It has to plug in somewhere

Twisted pair copper cabling

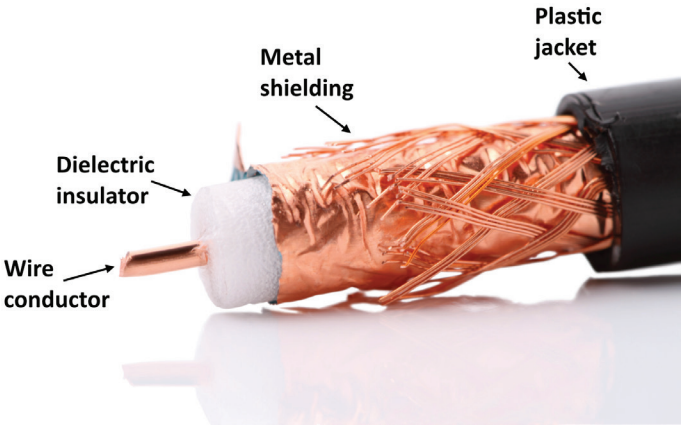
- Balanced pair operation
  - Two wires with equal and opposite signals
  - Transmit+, Transmit- / Receive+, Receive-
- The twist is the secret!
  - Keeps a single wire constantly moving away from the interference
  - The opposite signals are compared on the other end
- Pairs in the same cable have different twist rates

Cable categories

- Cables don't have a speed - the copper just sits there
- Electrical signals are sent over copper cable
  - The signal encoding determines the data transfer rate
- A cable must be manufactured to specific standards
  - IEEE 802.3 Ethernet standards determine the cable type
- Cable standards are described as a "category" of cable
  - Category 5e, Category 6A, etc.
  - Check the IEEE standard to determine the minimum cable category
  - The minimum cable category for 1000BASE-T is Category 5

Coaxial cables

- Two or more forms share a common axis
  - The inner conductor and the outer shield
- RG-6 used in television/digital cable
  - And high-speed Internet over cable



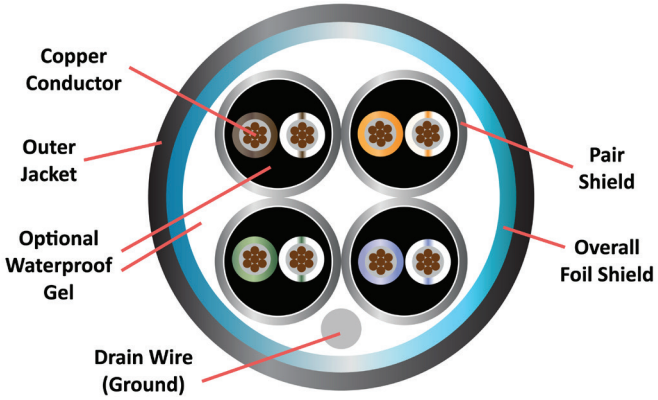
Unshielded and shielded cable

- UTP (Unshielded Twisted Pair)
  - No additional shielding
  - The most common twisted pair cabling
- STP (Shielded Twisted Pair)
  - Additional shielding protects against interference
  - Shield each pair and/or the overall cable
  - Requires the cable to be grounded
- Abbreviations
  - U = Unshielded
  - S = Braided shielding
  - F = Foil shielding
- (Overall cable) / (individual pairs)TP
  - Braided shielding around the entire cable and foil around the pairs is S/FTP
  - Foil around the cable and no shielding around the pairs is F/UTP

Ethernet Standard	Cable Category	Maximum Supported Distance
1000BASE-T	Category 5	100 meters
1000BASE-T	Category 5e (enhanced)	100 meters
10GBASE-T	Category 6	Unshielded: 55 meters Shielded: 100 meters
10GBASE-T	Category 6A (augmented)	100 meters

Direct burial STP

- Overhead cable isn't always a good option
  - Put the cable in the ground
- Provides protection from the elements
  - Designed to be waterproof
  - Often filled with gel to repel water
  - Conduit may not be needed
- Shielded twisted pair
  - Provides grounding, adds strength
  - Protects against signal interference



Plenum-rated cable

- Traditional cable jacket - Polyvinyl chloride (PVC)
- Fire-rated cable jacket
  - Fluorinated ethylene polymer (FEP) or low-smoke polyvinyl chloride (PVC)
- Plenum-rated cable may not be as flexible
  - May not have the same bend radius
- Worst-case planning
  - Important concerns for any structure

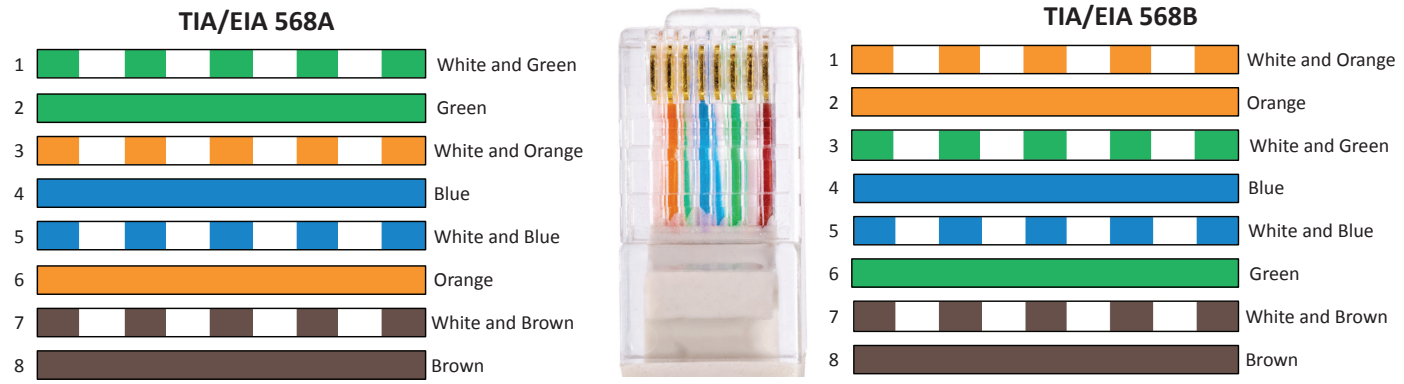
3.2 - 568A and 568B Colors

Structured cabling standards

- International ISO/IEC 11801 cabling standards
  - Defines classes of networking standards
- Telecommunications Industry Association (TIA)
  - Standards, market analysis, trade shows, government affairs, etc.
  - ANSI/TIA-568: Commercial Building Telecommunications Cabling Standard
  - http://www.tiaonline.org
- Commonly referenced for pin and pair assignments of eight-conductor 100-ohm balanced twisted pair cabling
  - T568A and T568B

T568A and T568B termination

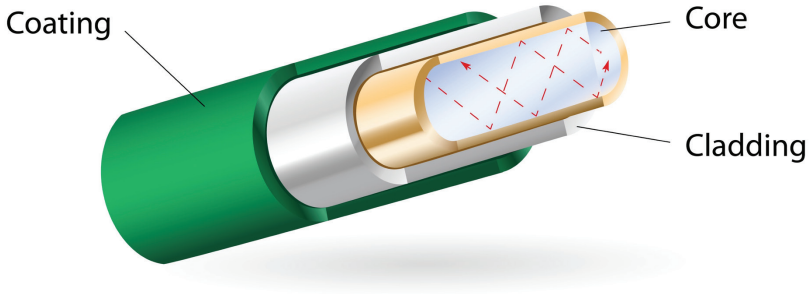
- Pin assignments from T568-B standard
  - Eight conductor 100-ohm balanced twisted-pair cabling
- T568A and T568B are different pin assignments for 8P8C connectors
  - Assigns the T568A pin-out to horizontal cabling
- Many organizations traditionally use 568B
  - Difficult to change in mid-stream
- You can't terminate one side of the cable with 568A and the other with 568B
  - This has never been the definition of a Gigabit Ethernet crossover cable



3.2 - Optical Fiber

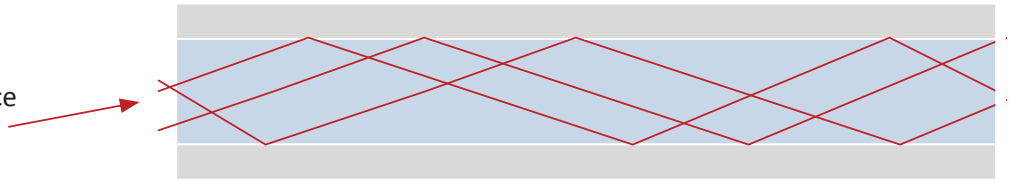
Fiber communication

- Transmission by light
  - The visible spectrum
- No RF signal
  - Very difficult to monitor or tap
- Signal slow to degrade
  - Transmission over long distances
- Immune to radio interference
  - There's no RF



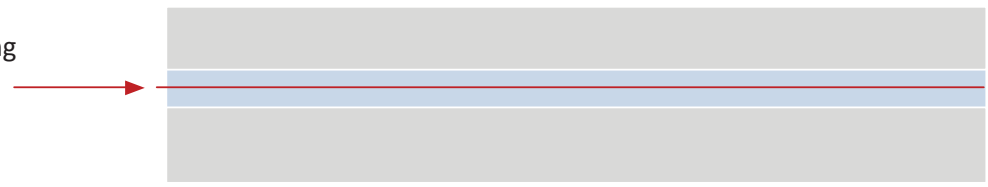
Multimode fiber

- Short-range communication
  - Up to 2 km
- Relatively inexpensive light source
  - i.e., LED



Single-mode fiber

- Long-range communication
  - Up to 100 km without processing
- Expensive light source
  - Commonly uses lasers



3.2 - Peripheral Cables

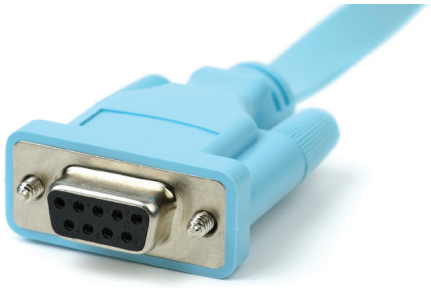
USB (Universal Serial Bus)
- Simplify connections - Printers, storage devices, keyboard, mouse
- USB 1.1
  - Low speed: 1.5 megabits per second, 3 meters
  - Full speed: 12 megabits per second, 5 meters
- USB 2.0 - 480 megabits per second, 5 meters
- USB 3.0 - SuperSpeed
  - 5 gigabits per second, ~3 meters
  - Standard does not specify a cable length
- USB 3.1
  - Maximum of 10 gigabits per second
- USB 3.2
  - Maximum of 20 gigabits per second

USB-C
- USB has a lot of different connectors
  - And they have changed over time
- Can be annoying to connect USB-A
  - Third time's the charm
- USB-C replaces all of these
  - One connector to rule them all
- USB-C describes the physical connector
  - It doesn't describe the signal

Standard-A Plug | **Standard-B Plug** | **Mini-B Plug** | **Micro-B Plug**

USB 3.0 Standard-B Plug | **USB 3.0 Standard-A Plug** | **USB 3.0 Micro-B Plug**

Serial cables
- D-subminiature or D-sub
  - The letter refers to the connector size
- Commonly used for RS-232
  - Recommended Standard 232
  - An industry standard since 1969
- Serial communications standard
  - Built for modem communication
  - Used for modems, printers, mice, networking
- Now used as a configuration port
- Directly connect to the device
  - Traditionally a serial connection
  - DB9 connector, RJ45 serial, USB connection



DB-9 / RS-232 Connector

- When all else fails - The console will be available
- A text-based serial interface - The console
- Requires a serial or USB connection
  - May need a USB to DB9 serial adapter

Thunderbolt
- High-speed serial connector
  - Data and power on the same cable
- Thunderbolt 1
  - Two channels
  - 10 Gbit/s per channel, 20 Gbit/s total throughput
  - Mini DisplayPort connector
- Thunderbolt 2
  - 20 Gbit/s aggregated channels, Mini DisplayPort connector
- Thunderbolt 3
  - 40 Gbit/s aggregated throughput
  - USB-C connector
- Maximum 3 meters (copper)
  - 60 meters (optical), Daisy-chain up to 6 devices
- Thunderbolt 4
  - Still 40 Gbit/s aggregated throughput
  - Support for dual 4K displays, Increased PCIe bandwidth

3.2 - Video Cables

HDMI (High-Definition Multimedia Interface)
- Video and audio stream
  - All digital, no analog
  - ~ 20 meter distance before losing too much signal
- 19-pin (Type A) connector
  - Proprietary connector

DisplayPort
- Digital information sent in packetized form
  - Like Ethernet and PCI Express
  - Carries both audio and video
- Compatible with HDMI and DVI
  - Passive adapter
  - DisplayPort -> HDMI
  - DisplayPort -> DVI
- Full-size interface may include a locking mechanism
  - Don't pull without unlocking

DVI (Digital Visual Interface)
- Single and dual link
  - Single link; 3.7 Gbps (HDTV at 60 fps)
  - Dual link; 7.4 Gbps (HDTV at 85 fps)
  - No audio support
- DVI-A - Analog signals
- DVI-D - Digital signals
- DVI-I - Integrated
  - Digital and analog in the same connector

VGA (Video Graphics Array)
- DB-15 connector
  - More accurately called DE-15
- Blue color - PC System Design Guide
- Video only - No audio signal
- Analog signal
  - No digital - Image degrades after 5 to 10 meters

USB-C
- A connector - Not a protocol
- Many uses for USB-C
  - Power, USB data, Thunderbolt data. DisplayPort video, HDMI video (not common), Mobile High-Definition Link (MHL)

HDMI

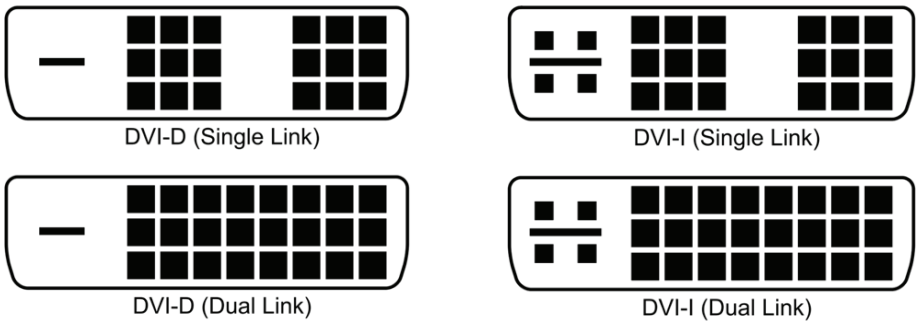
DisplayPort

Mini DisplayPort

VGA

DVI (Digital Visual Interface)

DVI (Digital Visual Interface)

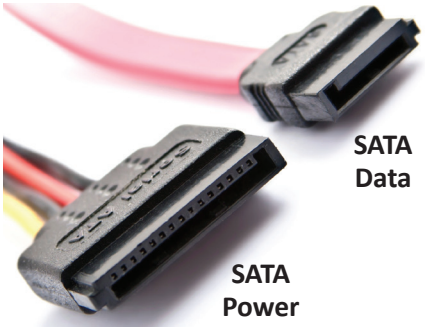


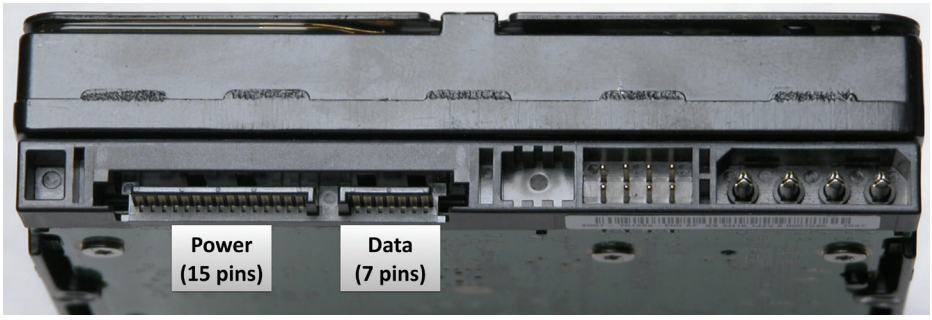
3.2 - Storage Cables

SATA (Serial AT Attachment)
- SATA Revision 1.0 - SATA 1.5 Gbit/s, 1 meter
- SATA Revision 2.0 - SATA 3.0 Gbit/s, 1 meter
- SATA Revision 3.0 - SATA 6.0 Gbit/s, 1 meter
- SATA Revision 3.2 - SATA 16 Gbit/s, 1 meter
- eSATA (external SATA) - Matches the SATA version, 2 meters

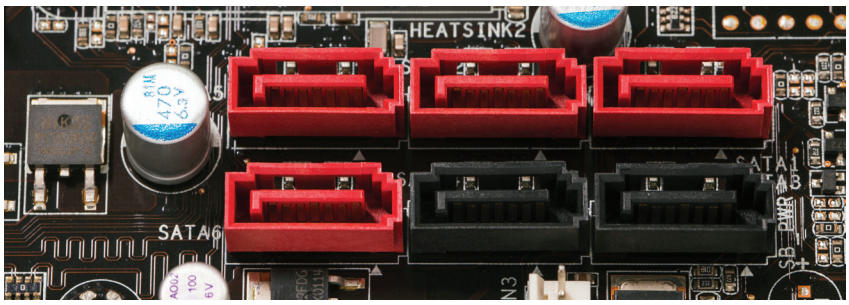
SATA
- One power cable and data cable per device
  - One-to-one

eSATA cable
- External device connections - Uses the SATA standard
- Similar in size to SATA
  - Connectors are physically different

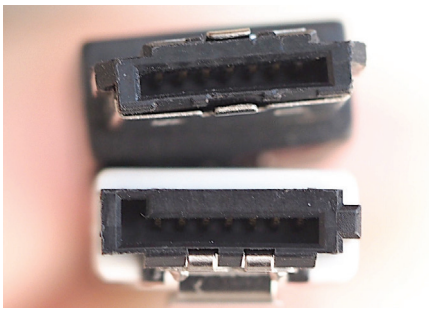




SATA data connectors on a motherboard



eSATA and SATA connectors



3.2 - Adapters and Converters

Adapters and converters
- The best laid plans…
  - Need an adapter
- Convert between different connectors
  - Electrically compatible
- Convert from one format to another
  - You need Ethernet but you only have USB
- A good temporary fix
  - Or a good permanent one

DVI to HDMI
- DVI-D and HDMI are electrically compatible
  - HDMI is backward-compatible with DVI-D
  - No signal conversion required
  - No loss of video quality

DVI to VGA
- DVI-A includes analog signals
  - Backward compatible with VGA
  - Only 640 x 480 is officially supported
- May only need an adapter - Analog to analog
- VGA to DVI digital will need a converter
  - Check your interface specifications

USB to Ethernet
- Some laptops don't have an Ethernet connection

USB-C to USB-A
- Merge the new with the old- Use your older peripherals

USB hub
  - Connect many devices - High speed USB connectivity

DVI-A

VGA

USB to Ethernet adapter

3.2 - Copper Connectors

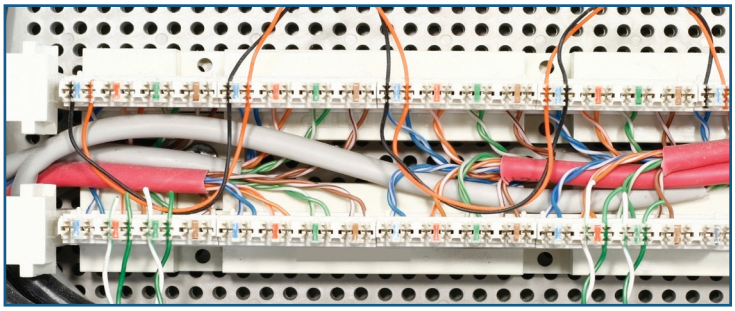
RJ11 connector
- 6 position, 2 conductor (6P2C)
  – Some cables will wire additional conductors
- Telephone or DSL connection

RJ45 connector
- Registered Jack type 45
- 8 position, 8 conductor (8P8C)
  – Modular connector
  – Ethernet

F-connector
- Cable television
  – Cable modem
  – DOCSIS
    (Data Over Cable Service Interface Specification)
- RG-6 cable - Threaded connector

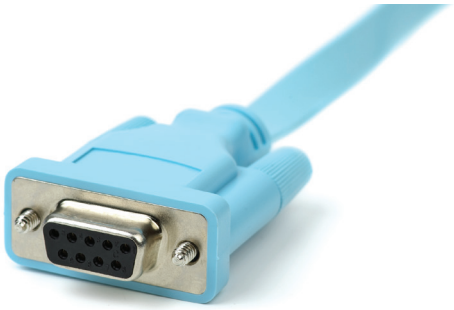
Punchdown block
- Wire-to-wire patch panel
  – No intermediate interface required
- Wires are “punched” into the block
  – Connecting block is on top
- Additional wires punched into connecting block
  – Patch the top to the bottom

110 block

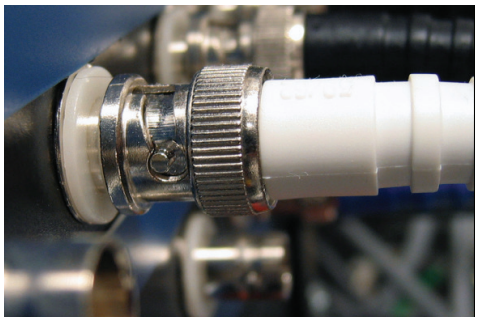


RJ11

RJ45

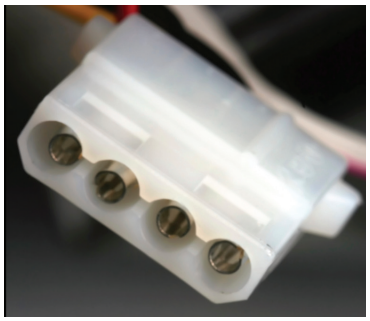


DB-9 / RS-232 Connector

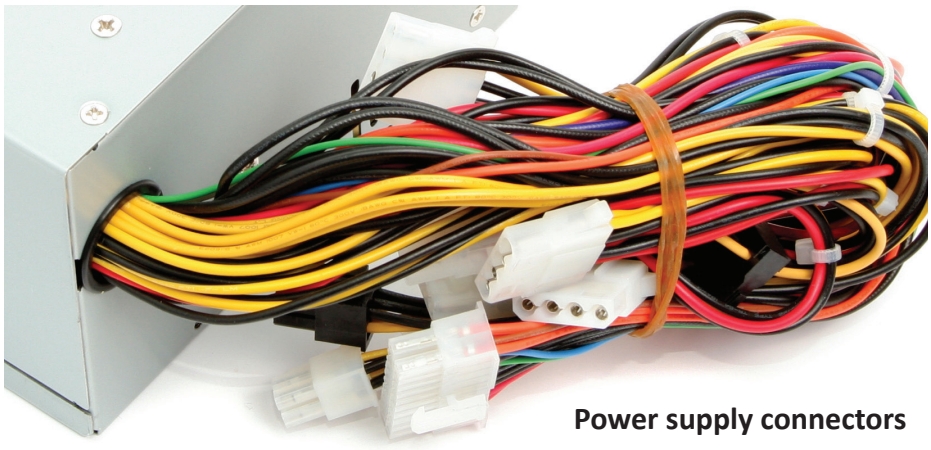


BNC connector

F-connector



Molex connector



Power supply connectors

3.2 - Copper Connectors (continued)

USB-C
- 24-pin double-sided USB connector
  - Used for both hosts and devices
- Used for USB, Thunderbolt
  - Interface is the same, signal can vary

Molex connector
- 4-pin peripheral power connector
  - Molex Connector Company
  - AMP MATE-N-LOK
  - Provides +12 V and +5 V
- Power for many devices in the computer case
  - Storage devices
  - Optical drives
  - Fans
  - Other peripherals

Lightning
- Apple proprietary
  - 8-pin digital signals
  - iPhone, iPad, iPod devices
- Some advantages over Micro-USB
  - Higher power output for phones and tablets
  - Can be inserted either way

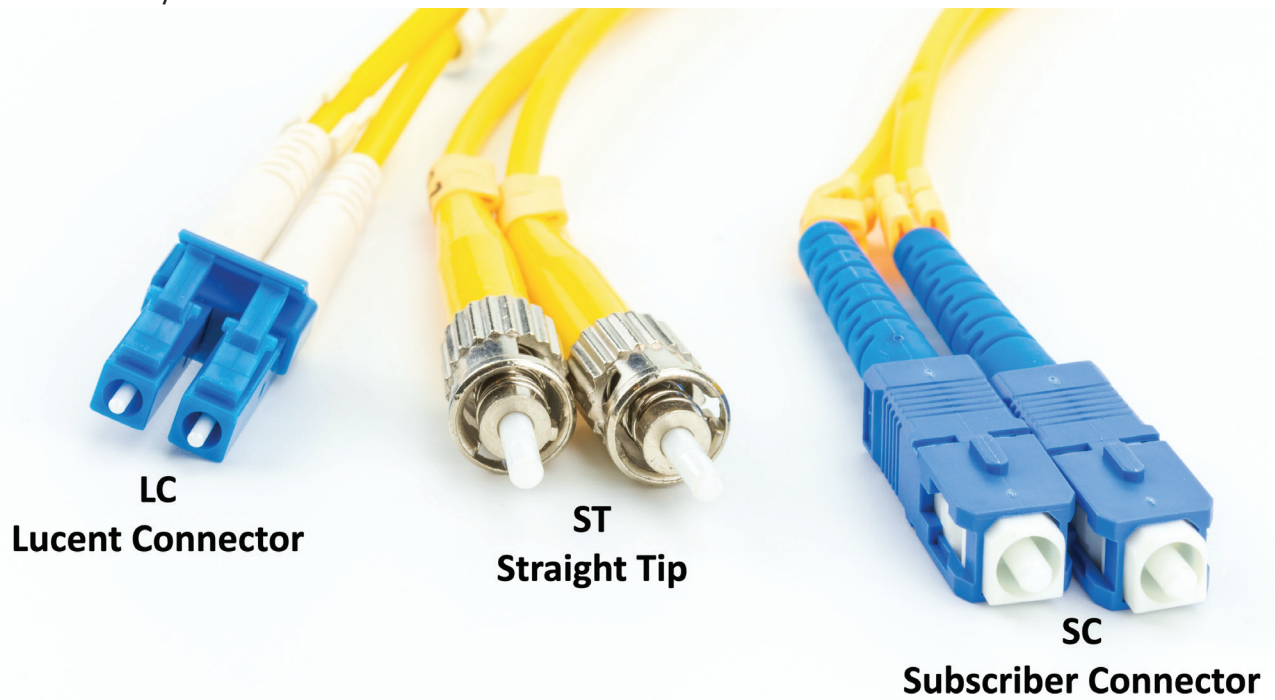
DB-9
- D-subminiature or D-sub
  - The letter refers to the connector size
- Commonly used for RS-232
  - Recommended Standard 232
  - An industry standard since 1969
- Serial communications standard
  - Built for modem communication
  - Used for modems, printers, mice, networking
- Now used as a configuration port
  - Management or Console port

3.2 - Fiber Connectors

ST - Straight Tip
- Bayonet connector - Stick and Twist
- Push on and turn
  - Locks in place. Turn to unlock.

SC - Subscriber Connector
- Not actually an abbreviation
  - We've created our own names
  - Square Connector or Standard Connector
- Pushes on to lock - Pull connector to unlock
- A popular fiber connector
  - Common in many data centers

LC - Lucent Connector
- Another popular fiber type
  - Smaller and more compact connector
- Locks in place with a clip
  - Press to release
- Other names
  - Local Connector
  - Little Connector



3.3 - An Overview of Memory

What is memory?
- Random Access Memory (RAM) is the most common
  – But it's not the only kind of memory
- RAM is not referring to hard drive or SSD storage
  – Don't mix the two terms
  – Data is stored permanently on the drive
- Data and programs can only be used when moved to RAM

RAM (Random Access Memory) slots
- Memory types have changed through the years
  – Driven by standardization and technology
- One of the most important components of your computer
  – Speed, speed, speed
- Every motherboard is different
  – Check your documentation

DIMM
- DIMM - Dual inline memory module
  – Electrical contacts are different on each side
- 64 bit data width

SO-DIMM
- Small Outline Dual In-line Memory Module
  – About half the width as a DIMM
- Used in laptops and mobile devices

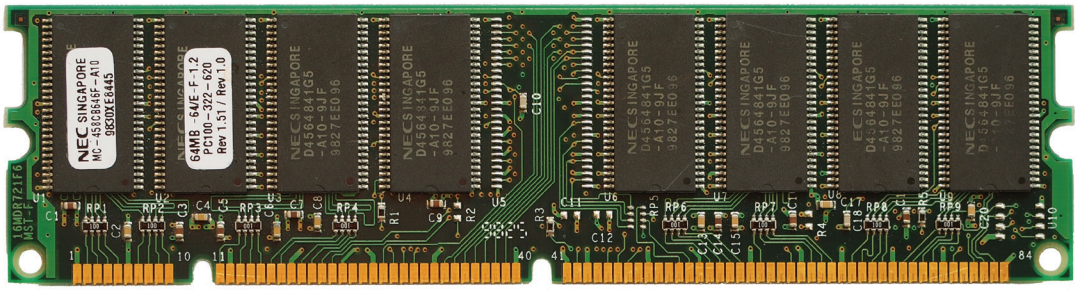
Dynamic Random Access Memory (DRAM)
- The memory on the DIMM
- Dynamic
  – Needs constant refreshing
  – Without refreshing, the data in memory disappears
- Random access
  – Any storage location can be accessed directly
  – Unlike magnetic tape

SDRAM
- Synchronous DRAM (SDRAM)
- SDRAM is synchronous with the common system clock
  – Queue up one process while waiting for another
  – Classic DRAM didn't wait for a clock signal

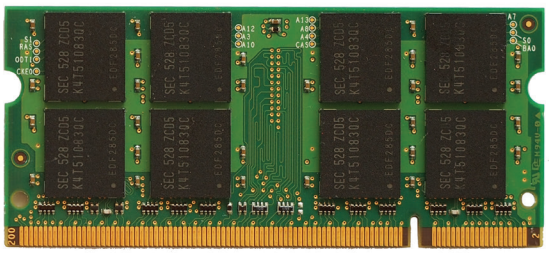
Double Data Rate 3 (DDR3) SDRAM
- Twice the data rate of DDR2
  – Larger chip capacities - Maximum 16 GB per DIMM
- No backwards compatibility
  – Speed brings sacrifice

Double Data Rate 4 (DDR4) SDRAM
- Speed increases over DDR3
  – Faster frequencies
  – Maximum 64 GB per DIMM
- Again, no backwards compatibility

Double Data Rate 5 (DDR5) SDRAM
- Faster data transfers between the memory module and motherboard
  – Maximum of 64 GB per DIMM
- The key has moved - No backwards compatibility



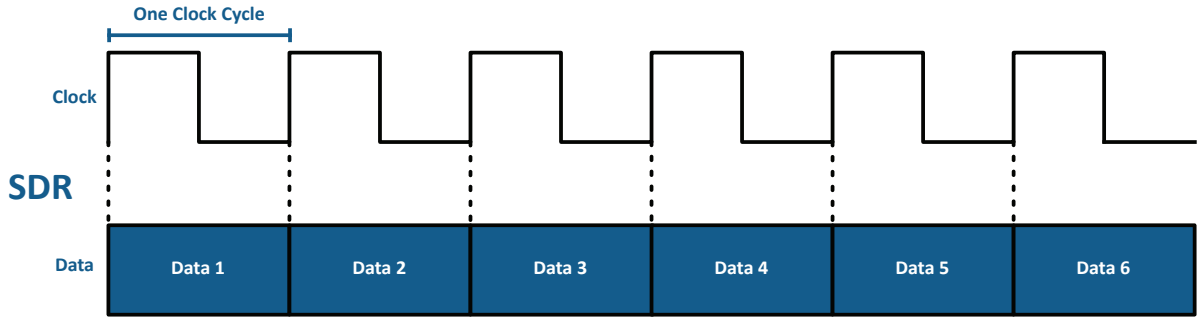
DIMM
(Dual Inline Memory Module)

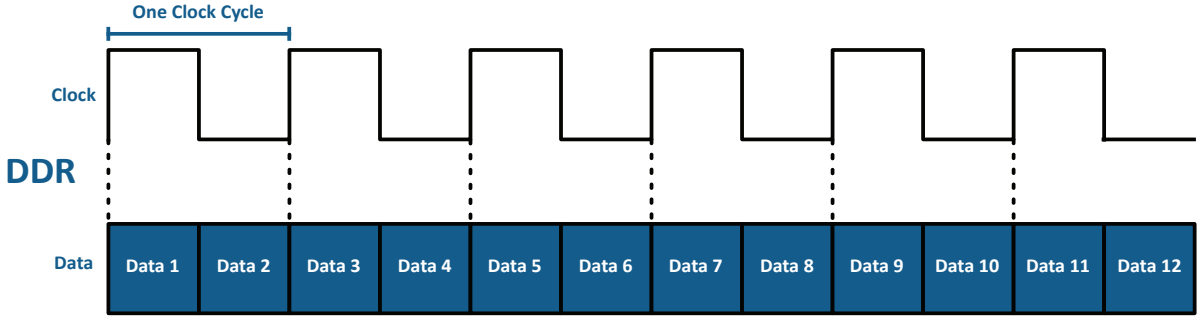


SO-DIMM
(Small Outline Dual Inline Memory Module)

3.3 - An Overview of Memory (continued)

SDR vs. DDR





3.3 - Memory Technologies

Memory that checks itself

- Used on critical computer systems
  - VM servers, database servers, any server
- Parity memory
  - Adds an additional parity bit
  - Won't always detect an error
  - Can't correct an error
- Error Correcting Code (ECC)
  - Detects errors and corrects on the fly
  - Not all systems use ECC
  - It looks the same as non-ECC memory

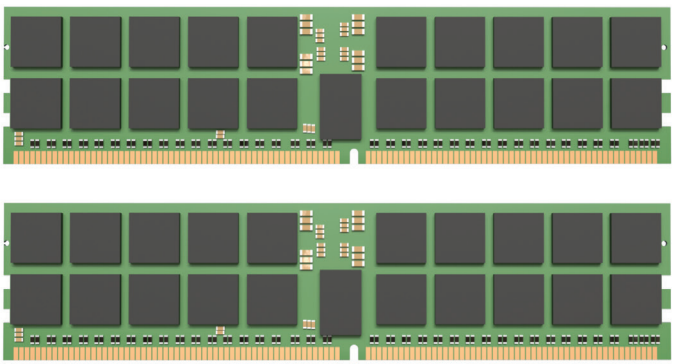
CPU to RAM throughput

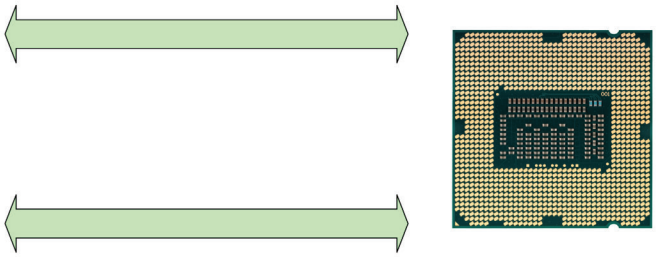
- There's a maximum throughput between RAM and CPU
  - The memory bandwidth
- Measured in MT/s
  - Megatransfers/s - Million transfers per second
  - e.g., 32 GB DDR5, 1x32GB, 5600MT/s
- Faster is better - Physics becomes the challenge

Multi-channel memory

- Difficult to increase the speed
  - Push as much as possible across a single memory bus
- Two independent memory buses
  - Doubles the throughput
  - Two memory modules are faster than one
- Dual-channel, triple-channel, or quad-channel
- Memory combinations should match
  - Exact matches are best
- Memory module slots are often colored differently

Multi-channel memory





3.4 - Storage Devices

Storage devices
- Need some way to store information when the computer is off
  - RAM requires power to maintain the data
- Many different ways to store data
  - Hard drives
  - Solid-state drives
  - Flash drives
  - Memory cards
  - Optical drives

Hard disk drives (HDD)
- Non-volatile magnetic storage - rapidly rotating platters
- Random-access
  - Retrieve data from any part of the drive at any time
- Moving parts
  - Spinning platters, moving actuator arm
  - Mechanical components limit the access speed
  - Mechanical components can also break

Solid-state drives (SSD)
- Non-volatile memory - No moving parts
- Very fast performance - No spinning drive delays

PCIe storage interfaces
- High speed for SSD communication - Used before M.2
- Connect directly to the motherboard
  - Provides power, high throughput
- PCI Express
  - Modern PCIe standards can transfer data at 64 gigabits per second per lane

NVMe
- SATA was designed for hard drives
  - Uses AHCI (Advanced Host Controller Interface) to move drive data to RAM
  - SATA revision 3 throughput up to 600 MB/s
  - SSDs need a faster communication method
- NVMe (Non-volatile Memory Express)
  - Designed for SSD speeds
  - Lower latency, supports higher throughputs
  - Uses the PCI Express bus
  - Often connects with an M.2 interface

Flash memory

Serial attached SCSI
- The latest generation of SCSI technology
  - Increased storage throughput
  - Serial communication
  - SCSI protocol
  - SAS - Serial Attached SCSI
- The control and management of SCSI
  - Useful in large storage arrays
- An upgrade to SATA throughput
  - SATA can communicate at 6 gigabits per second
  - High-end SAS can reach 22.5 gigabits per second

mSATA (Mini-SATA)
- Shrink the SATA drive into smaller devices
  - Same data, different form factor
  - Great for laptops and mobile devices
- Smaller than 2.5” SATA drives
  - No spinning drive - Allows for different form factors
- Was used briefly - Quickly replaced by the m.2

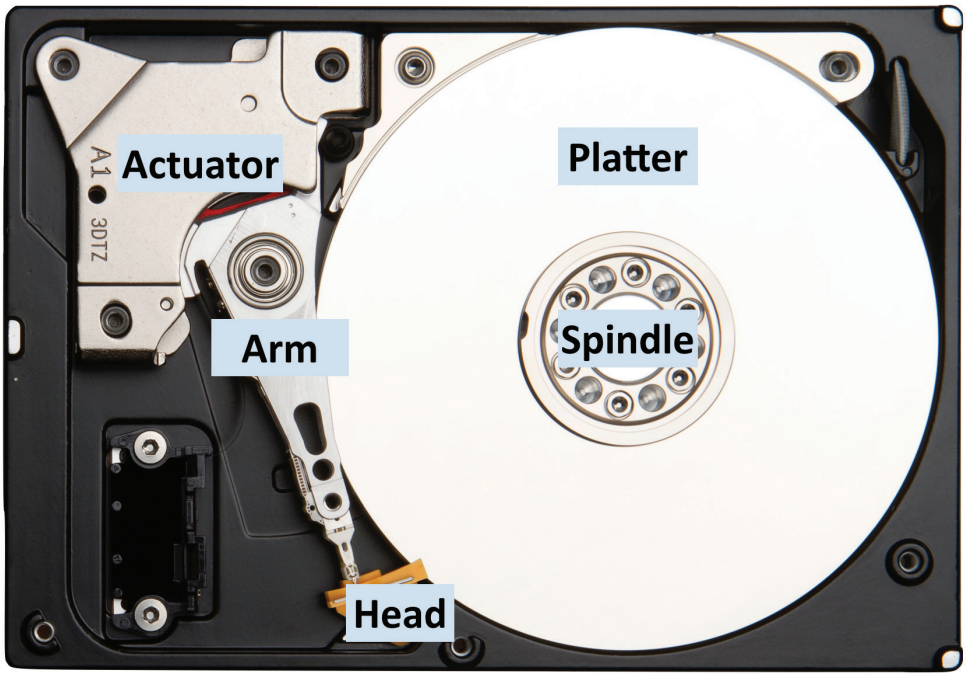
M.2 interface
- Smaller form factor - No SATA data or power cables
- Can use a PCI Express bus connection
  - 4 GB/second throughput or faster when using NVMe PCIe x4
- Different connector types
  - Needs to be compatible with the slot key/spacer
  - B key, M key, or B and M key
  - Some M.2 drives will support both

B-key and M-key
- M.2 doesn’t guarantee NVMe
  - Your M.2 interface may be using AHCI
  - Check your documentation
- Your motherboard may only support one type of M.2 key
  - Check the documentation again

Flash drives
- Flash memory
  - EEPROM (Electrically erasable programmable read-only memory)
  - Non-volatile memory - No power required to retain data
- Limited number of writes
  - Can still read the data
- Not designed for archival storage
  - Easy to lose or damage - Always have a backup
  - Flash memory

Optical drives
- Small bumps read with a laser beam
  - Microscopic binary storage
- Relatively slow - Archival media
- Many different formats - CD-ROM, DVD-ROM, Blu-ray
- Internal and external drives
  - For those uncommon application installations

3.4 - Storage Devices (continued)



Rotational Speed (rpm)	Average Rotational Latency
15,000	2 ms
10,000	3 ms
7,200	4.16 ms
5,400	5.55 ms

3.4 - RAID

Data redundancy

- Hard drives store huge amounts of important data
- Hard drives are moving components - They will eventually break
- What happens to the data when the drive fails?
  – You can prepare for that; Use an array of drives
- RAID is not backup

RAID

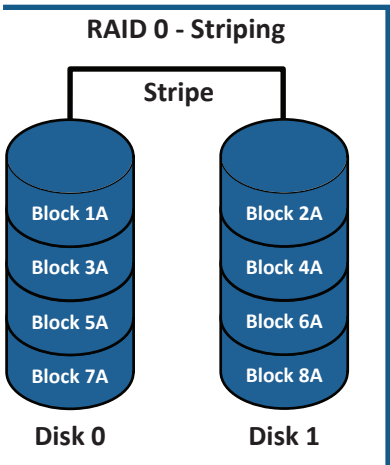
- Redundant Array of Independent Disks
  – They're also inexpensive disks.
- Different RAID levels - Some redundant, some not
- RAID 0 – Striping
- RAID 1 - Mirroring
- RAID 5 – Striping with one parity drive
- RAID 6 - Striping with two parity drives
- Nested RAID - RAID 1+0 (a.k.a. RAID 10) - A stripe of mirrors

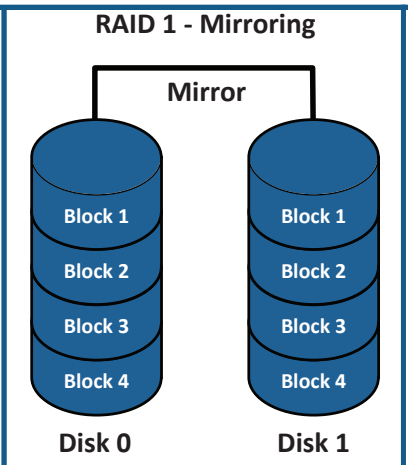
RAID 0 - Striping

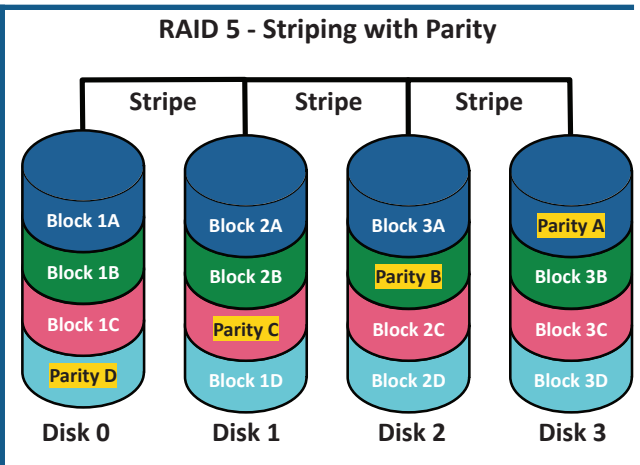
- File blocks are split between two or more physical drives
  – High performance
  – Data written quickly
- No redundancy
  – A drive failure breaks the array
  – Raid 0 is zero redundancy

RAID 1 - Mirroring

- File blocks are duplicated between two or more physical drives
- High disk utilization
  – Every file is duplicated
  – Required disk space is doubled
- High redundancy
  – Drive failure does not affect data availability





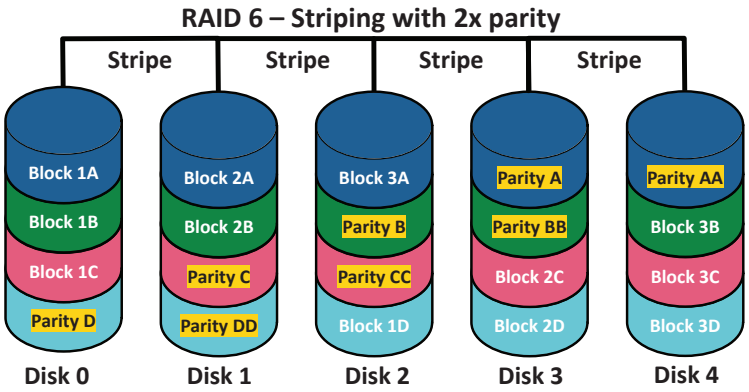


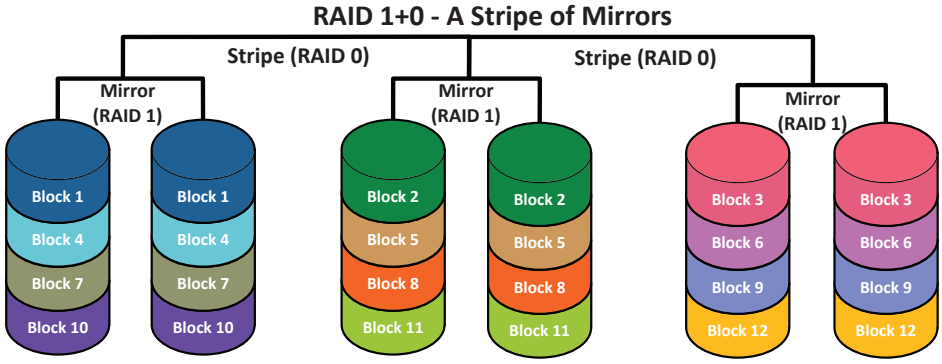
3.4 - RAID (continued)

RAID 5 - Striping with parity
- File blocks are striped
  - Along with a parity block
  - Requires at least three disks
- Efficient use of disk space
  - Files aren't duplicated, but space is still used for parity
- High redundancy
  - Drive is available after a failure
  - Parity calculation may affect performance

RAID 6
- Add another parity block
  - Requires at least four drives
- Could lose two drives
  - Data would continue to be available
- Requires another drive
  - Adds more parity
  - Does not add more capacity

RAID 1+0 - A stripe of mirrors
- The speed of striping, the redundancy of mirrors
- Need at least 4 drives





3.5 - Motherboard Form Factors

Motherboard form factors
- Physical size
  - Case sizing
- Basic layout
  - Room for small changes
- Power
  - Standard connectors
- Airflow
  - Increasingly important
- Wikipedia: There are 40 different motherboard types categorized!

What you need to know
- CompTIA exam objectives:
  - "3.5 - Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards."
- Focus on which motherboard would be the best choice for a scenario
  - Memorizing the dimensions of the motherboard probably isn't the most important thing
  - ATX, microATX, or ITX
- Which motherboard would you choose?
  - Media center, video editing, thin client, etc.

ATX form factor
- Advanced Technology Extended
  - Standardized by Intel in 1995
  - Seen many updates over the years
- Power
  - 20 pin connector
  - 24 pin connector, additional 4/8 pin connector

microATX (uATX, M-ATX)
- Smaller than an ATX motherboard
  - Limited expansion slots
- Backward compatibility
  - Similar mounting points
  - Similar power
- Very popular form factor
  - Many different available options

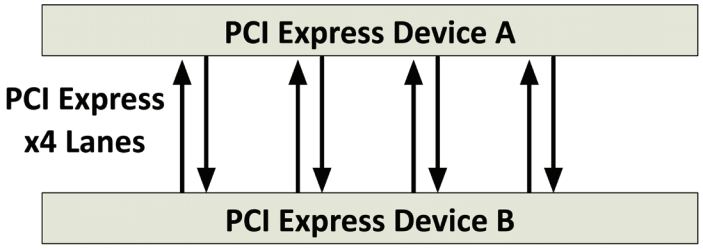
ITX form factor
- A series of low-power motherboards
  - Developed by VIA Technologies in 2001
  - Small form factor
- Mini-ITX is screw-compatible with ATX
  - Fits almost any enclosure
- Small form factor uses
  - Single-purpose computing, i.e., streaming media

3.5 - Motherboard Expansion Slots

A computer bus
- Communication path - A city of technology
- Internal PC growth - Independent pathways
- System expansion - Additional capabilities

PCI
- Peripheral Component Interconnect
  – Created in 1994
- Many expansion options
  – 32-bit and 64-bit bus width
  – Parallel communication
- A common expansion interface on previous computer generations
  – PCI Express is the newer technology

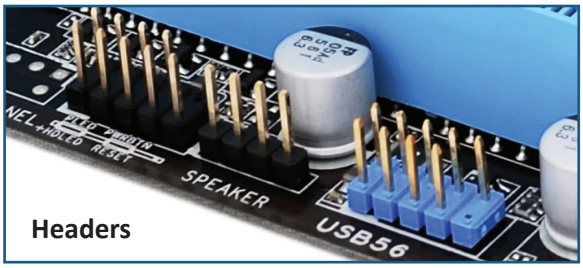
PCI Express
- Also known as PCIe - Replaces the older PCI standard
- Communicates serially - Unidirectional serial "lanes"
  – Slower devices don't slow down everyone
- One, two, four, eight, sixteen, or thirty-two full-duplex lanes
  – x1, x2, x4, x8, x16, x32
  – "x" is pronounced "by" ("by 4," "by 16")



3.5 - Motherboard Connections

24-pin motherboard power
- Main motherboard power
  – Provides +3.3 V, +/-5 V, and +/- 12 V
- 20 pin connector was the original ATX standard
  – 24 pin was added for PCI Express power
- You can connect a 24-pin connector to a 20-pin motherboard
  – Some cables are 20-pin + 4-pin

PCIe 6-pin and 8-pin power
- Additional power for PCI Express adapters
  – 6-pin (75 watts), 8-pin (150 watts)
  – Provides +12 V
- Commonly seen on high-power components
  – Video adapters
- Some cables have two pieces
  – 6 pin and 2 pin = 8 pin
  – Use on both 6 pin and 8 pin connectors



Headers

3.5 - Motherboard Compatibility

Intel and AMD
- Two solid CPU manufacturers
  – The differences are subtle
- Cost
  – AMD tends to be a bit less expensive
- Different sockets
  – The motherboard is designed for a particular CPU
- Choosing a side
  – Historically, AMD is value and Intel is performance
  – These switch places often

Server motherboards
- Multisocket
  – Supports multiple physical CPU packages
  – Split the load
- Memory slots - Supports 4+ modules
- Expansion slots
  – Many slots and different sizes
- Overall size
  – Designed for a rack-mounted system
  – Larger ATX-sized systems

3.5 - The BIOS

BIOS - Basic Input/Output System
- The software used to start your computer
  - The firmware - System BIOS, ROM BIOS
  - ROM or flash memory
- Initializes CPU and memory - Build the workspace
- POST - Power-On Self-Test
- Look for a boot loader - Start the operating system

Legacy BIOS
- The original / traditional BIOS
  - It's been around for more than 25 years
- Older operating systems talked to hardware through the BIOS
  - Instead of accessing hardware directly
- Limited hardware support
  - No drivers for modern network, video, and storage devices

UEFI BIOS
- Unified Extensible Firmware Interface
  - Based on Intel's EFI (Extensible Firmware Interface)
- A defined standard - Implemented by the manufacturers
- Designed to replace the legacy BIOS
  - Need a modern BIOS for modern computers
  - Graphical and text-based

3.5 - BIOS Settings

The secret button(s)
- Launching the system setup
  - Del, F1, F2, Ctrl-S, Ctrl-Alt-S
- Hyper-V (Windows 8/10/11)
  - Turn Windows features on and off
- VMware Workstation Player
  - http://www.vmware.com/products/player/
- But not VirtualBox
  - http://www.virtualbox.org
- Many simulators online
  - Search for "UEFI BIOS simulator"

Fast startup
- Windows 10 and 11 (and previous versions)
  - Doesn't actually shut down all the way
  - Starts up so quickly, you can't open the BIOS configuration
- From the Windows desktop
  - Hold down shift when clicking Restart
  - Settings / Update & Security / Recovery / Advanced startup / Restart now
  - System Configuration (msconfig)
- Interrupt normal boot three times
  - Presents the boot option screen

Important tips
- Make a backup of your BIOS configuration
  - Write detailed notes
  - Take a picture
- Don't make a change unless you're certain of the setting
- It's difficult to leave it alone
- Did I mention the backup?
  - It can save a lot of time!

Boot options
- What happens when you power on?
  - The BIOS knows
- Disable hardware
  - Unavailable to the operating system
- Modify the boot order
  - Which boots first?
  - Move to the next in order
  - USB drive, SSD, hard drive

USB permissions
- A security challenge
  - Very small storage devices
  - Very large capacities
- USB connections
  - Convenient
  - High speed
- US Department of Defense banned USB flash media for 15 months in 2008
  - SillyFDC worm

Fans
- Computing power creates heat
  - A lot of heat
- Many different cooling fans
  - CPU fan
  - Chassis fans
- Motherboards often include an integrated fan controller
  - And temperature sensors
  - The motherboard increases and decreases the fan speeds

3.5 - BIOS Settings (continued)

Secure Boot
- Malicious software can "own" your system
  - Malicious drivers or OS software
- Secure boot
  - Part of the UEFI specification
- Digitally sign known-good software
  - Cryptographically secure
  - Software won't run without the proper signature
- Support in many different operating systems
  - Windows and Linux support

UEFI BIOS Secure Boot
- UEFI BIOS protections
  - BIOS includes the manufacturer's public key
  - Digital signature is checked during a BIOS update
  - BIOS prevents unauthorized writes to the flash
- Secure Boot verifies the bootloader
  - Checks the OS bootloader's digital signature
  - Bootloader must be signed with a trusted certificate
  - Or a manually approved digital signature

Boot password management
- BIOS Password / User Password
  - System won't start
  - Need the password to start the operating system
- Supervisor Password
  - Restrict BIOS changes
  - Must use supervisor password to change any BIOS configurations
- Remember your password!
  - Must reset the BIOS to recover
  - Usually a jumper

Clearing a boot password
- Store the BIOS configuration - Your settings
- Complementary metal-oxide semiconductor (CMOS)
  - A type of memory
  - May be backed up with a battery
- Usually flash memory these days
  - Easily stored and accessed
- Reset with a jumper
  - Short (connect) two pins on the motherboard

Temperature monitoring
- Part of the BIOS
  - Built-in diagnostics
- Run from the BIOS menu
  - No additional media or software required
- Focused on hardware checks
  - Doesn't touch the operating system

Virtualization support
- Run other OSes within a single hardware platform
  - Multiple operating systems share physical hardware components
- Virtualization in software was limited
  - Performance and hardware management challenges
- Virtualization added to the processor
  - Hardware is faster and easier to manage
  - Intel Virtualization Technology (VT)
  - AMD Virtualization (AMD-V)

3.5 - HSM and TPM

Keeping secrets
- IT security is based around secrets
  - And trusting other devices and applications
- We use a lot of encryption
  - Web servers, file storage, phone data, and more
- The secret part is the key
  - Unlocks all of this encryption
- We need to store this key somewhere
  - How do you securely store these secrets?
  - Where do you store these secrets?

Trusted Platform Module (TPM)
- A specification for cryptographic functions
  - Hardware to help with single device encryption functions
- Cryptographic processor
  - Random number generator, key generators
- Persistent memory
  - Includes unique keys burned in during production
- Versatile memory
  - Storage keys, hardware configuration information
- Password protected
  - No dictionary attacks
- Contains a unique secret key
  - Not available outside of this device
- Links this key to this computer
  - Can't move an encrypted drive to another computer
  - The key is on this computer
- Also used as a physical point of reference
  - A root of trust
  - Has this computer been modified or tampered with?
- Can't use this key on another computer
  - Cryptography for a single device

3.5 - HSM and TPM (continued)

Hardware Security Module (HSM)
- Often used in large environments
  - Server clusters, many diverse devices
- Key backup
  - Secured storage for servers
  - Lightweight HSMs for personal use
  - (Smart card, USB, flash memory)
- Often high-end cryptographic hardware
  - Plug-in card or separate hardware device
- Cryptographic accelerators
  - Perform cryptographic functions for other devices
  - Only the HSM knows the key

TPMs and HSMs
- TPM (Trusted Platform Module)
  - Used on a single system
  - Secure data on a local device
  - Often built into a motherboard or available as an add-on module
  - Mobile phone booting, screen locking, and encrypted storage
- HSM (Hardware Security Module)
  - Used by many systems
  - Secure data across multiple devices
  - Often deployed as a high-end server or appliance in a data center
  - Protect the Certificate Authority key on a central secure device

3.5 - CPU Features

Operating system technologies
- 32-bit vs. 64-bit - Processor specific
- 32-bit processors = 2^{32} = 4,294,967,296 values
- 64-bit processors = 2^{64} = 18,446,744,073,709,551,616 values
  - 4 GB vs. 17 billion GB
  - The OS has a maximum supported value

Operating system technologies
- Hardware drivers are specific to the OS version (32-bit / 64-bit)
  - 32-bit (x86), 64-bit (x64)
- 32-bit OS cannot run 64-bit apps
  - But 64-bit OS can run 32-bit apps
- Apps in a 64-bit Windows OS
  - 32-bit apps: \Program Files (x86)
  - 64-bit apps: \Program Files

Advanced RISC Machine (ARM)
- CPU architecture developed by Arm Ltd.
  - They design the chip, others license and build
- Simplified instruction set
  - Efficient and fast processing
  - Less power
  - Less heat
- Traditionally used for mobile and IoT devices
  - Adoption by Apple and other manufacturers for desktops and laptops
  - The lines are constantly blurring

Processor cores
- Dual-core / Quad-core / Octa-core / Multi-core
- Multiple cores
  - Each core has its own cache
  - The entire chip may have a shared cache

3.5 - Expansion Cards

Expansion cards
- Extend the functionality of your computer
  - You may need more than the motherboard provides
- Relatively simple process
  - Designed for end-user installation
- Install hardware - Add a card
- Install a driver - Software for the operating system

Sound card
- Output
  - High-end audio
  - Advanced headphone amp
  - Home theater, Dolby decoding
- Multiple inputs
  - Music capture, podcasting, microphone

Video card
- Many CPUs include an integrated GPU
  - Video functionality is built into the CPU package
- Discrete graphics - The GPU is not part of the CPU
- Separate interface card - High performance

Capture card
- Video as an input
  - Video capture, live streaming, external cameras
- High performance
  - Video bandwidths can be quite high
  - PCI Express connection

Network Interface Card (NIC)
- Ethernet connection
  - Motherboard may not include a NIC
  - Onboard NIC may not be working
- Additional connections - Servers, routers, security devices

3.5 - Expansion Cards (continued)

Documentation
- Check the motherboard documentation
  – Number and type of slots
- Check the adapter card documentation
  – Minimum requirements - Hardware, software
  – Knowledge base - Any “gotchas?”
  – Support forums - What are other people saying?
- You may need to perform a driver installation before installing hardware

Driver installation
- You checked the documentation first, right?
- Check the web for the latest version
- Uninstall previous drivers through Windows Device Manager
- Manufacturer provided setup program
- Manual installation through Windows Device Manager
- Check the Device Manager for the status - Does it work?

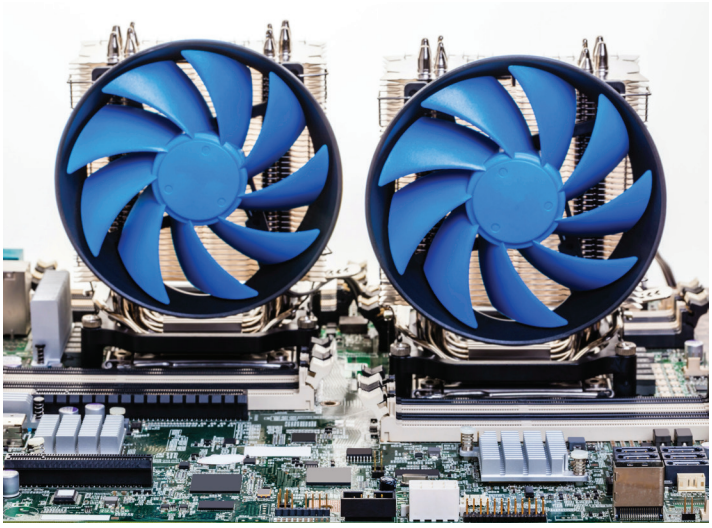
3.5 - Cooling

Case fans
- Cool air is pulled through a personal computer
  – Always check for good airflow
- Motherboard layout becomes important
- Component location is key
  – Devices, wiring, power
- Many different sizes and styles
  – And volume levels

On-board fans
- Designed to cool an entire adapter card
- Can be bulky
  – May take additional adapter card space
- Usually seen on high-end graphics cards

Fan specifications
- Standard sizes
  – 80 mm, 120 mm, 200 mm, etc.
- Different speeds
  – Variable speed
- Different noise levels
  – Not all fans sound the same

Heat sink and fans



Fanless / passive cooling
- No fans, no noise - Silent operation
- Specialized functions
  – Video servers, TV set top box, satellite receiver, media server
- Functions are very controlled
  – Carefully measured thermal tests
- Low-power components - Heat sinks

Heat sink
- Dissipate heat through thermal conduction
  – Copper or aluminum alloy
- Fins/grid increase surface area
  – Heat is then transferred to the cooler air
- They get HOT – don’t touch them!
- Thermal paste creates a good contact between the chip and the heat sink

Thermal paste
- Thermal grease, conductive grease
  – Thermally conductive adhesive
- Place between the heat sink and the component
  – Improves thermal conductivity
  – Moves the heat away from the component
- A little bit goes a long way
  – Pea-sized application

Thermal pad
- Conduct heat without the mess - Cut to size and install
- Easy to use - Won’t leak and damage components
- Almost as effective as thermal paste
  – But still very good
- Not reusable - Remove and replace

Liquid cooling
- Coolant is circulated through a computer
  – Not a new concept
  – Automobiles, mainframe computers
- High-end systems
- Gaming, graphics
- Overclocking

3.6 - Computer Power

WARNING

- **Always disconnect from the power source when working on a device**
  - Always. Seriously.
- Some devices store a charge in capacitors
  - Know how to discharge before touching
- **Never connect your body to any part of an electrical system**
  - **Do not connect yourself to the ground wire of an electrical system**
- **Respect electricity**
  - It does not respect you

Computer power supply

- Computer uses DC voltage
  - Most power sources provide AC voltage
- Convert 120 V AC or 240 V AC
  - To 3.3 V DC, 5 V DC, and 12 V DC
- You'll know when this isn't working
  - An important component

Amp and volt

- Ampere (amp, A) – The rate of electron flow past a point in one second
  - The diameter of the hose
- Voltage (volt, V)Electrical "pressure" pushing the electrons
  - How open the faucet is

Power

- Watt (W) – Measurement of real power use
  - volts * amps = watts
  - 120V * 0.5A = 60W

Current

- Alternating current (AC)
  - Direction of current constantly reverses
  - Distributes electricity efficiently over long distances
  - Frequency of this cycle is important
  - US/Canada – 110 to 120 volts of AC (VAC), 60 hertz (Hz)
  - Europe – 220-240 VAC, 50 Hz
- Direct current (DC)
  - Current moves in one direction with a constant voltage

Dual-voltage input options

- Voltage varies by country
  - US/Canada – 120 volts of AC (VAC), 60 hertz (Hz)
  - Europe – 230 VAC, 50 Hz
- Manually switch between 120 V and 230 V
  - Get your meter!
  - Or use an auto voltage power supply
- Don't plug a 120 V power supply into a 230 V power source!

Power supply output

- Different voltages
  - For different components
- Positive and negative voltage
  - Voltage is a difference in potential
  - The electrical ground is a common reference point
  - Depends on where you measure from
- At the front door of your house
  - The second floor is +10 feet
  - The basement is -10 feet
- +12 V
  - PCIe adapters, hard drive motors, cooling fans, most modern components
- +5 V
  - Some motherboard components
  - Many components are now using +3.3 V
- +3.3 V
  - M.2 slots, RAM slots, motherboard logic circuits
- +5 VSB
  - Standby voltage
- -12 V
  - Integrated LAN
  - Older serial ports
  - Some PCI cards
- -5 V
  - Available for ISA adapter cards
  - Most cards didn't use it
  - Today's motherboards don't have ISA slots

24-pin motherboard power

- Main motherboard power
  - Provides +3.3 V, +/-5 V, and +/- 12 V
- 20 pin connector was the original ATX standard
  - 24 pin was added for PCI Express power
- You can connect a 24-pin connector to a 20-pin motherboard
  - Some cables are 20-pin + 4-pin

Redundant power supplies

- Two (or more) power supplies
  - Internal to the server
- Each power supply can handle 100% of the load
  - Would normally run at 50% of the load
- Hot-swappable
  - Replace a faulty power supply without powering down

3.6 - Computer Power (continued)

Power supply connectors

- Fixed connectors
  - Connected to the power supply
  - May have too many connectors
  - May not have enough
- Modular
  - Add cables as needed
  - Fewer leftover wires,better airflow
  - A bit more expensive

Sizing a power supply

- Power supplies are rated by watts
  - Overall and by individual voltages
- Bigger isn't necessarily better
  - More expensive
  - Doesn't speed up your computer
- Physical size is relatively standard
  - Older cases and systems may have proprietary sizes
- Calculate the watts required for all components
  - CPU, storage devices, video adapter
  - Many online calculators
- Video adapters are usually the largest power draw
  - Many video card specifications list a recommended power supply wattage
- 50% capacity is a good rule of thumb
  - Power supply runs efficiently and there's room to grow

Energy efficiency

- Converting AC to DC isn't perfect
  - Some power is often lost
  - Heat is generated
- Check the efficiency range
  - Power supplies often range between 80% to 96%
  - Can depend on voltage and redundant power
- More efficiency means more DC power
  - And less heat; save money!
- There's an established certification program
  - Pick the right efficiency for your requirements Overall and by individual voltages

3.7 - Multifunction Devices

Multifunction devices

- Multifunction devices (MFD)
  - Printer
  - Scanner
  - Fax (yes, really)
  - Network connection
  - Phone line connection
  - Print from web
- A lot of things can go wrong
  - You're going to fix them

Unboxing a device

- These can be large devices
  - Make sure you have enough room
  - Stay out of walkways
- Check the area
  - Power
  - Network
  - Accessibility

Printer drivers

- Specific to a printer model
- Get this exactly right
- Get the right operating system drivers
  - Windows 10, Windows 11
  - Get the right version of the operating system
  - 32-bit drivers are very different than 64-bit drivers
- You can't mix and match drivers
  - It's a very specific task

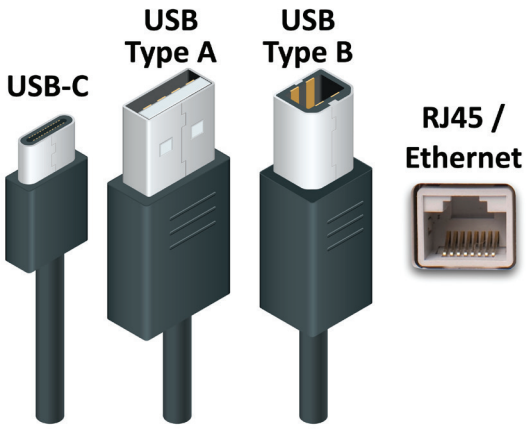
PCL or PostScript

- PCL (Printer Command Language)
  - Created by Hewlett-Packard
  - Commonly used across the industry
- PostScript
  - Created by Adobe Systems
  - Popular with high end printers
- Make sure the drivers match the printer
  - PCL printer, PCL driver
  - PostScript printer, PostScript driver
  - Wired device sharing

3.6 - Multifunction Devices (continued)

Firmware
- The internal "operating system" of the multifunction device
  - Starts the system
  - Connects to the network
  - Interprets the incoming data stream
  - Runs the printing process
- Update the firmware to avoid any incompatibilities
  - And probably fix a few bugs
- Get the latest firmware from the device website
  - Install from a stable platform and power source The most common connector

Wired device sharing
- USB
  - The most common connector
  - USB Type B on the printer, USB Type A on the computer
  - Or, USB-C everywhere
- Ethernet - RJ45 connector
- May include more than one option



Wireless device sharing
- Bluetooth - Limited range
- 802.11 Infrastructure mode
  - Many devices using an access point
- 802.11 Ad hoc mode
  - No access point
  - Direct link between wireless devices

Sharing the printer
- Printer share
  - Printer is connected to a computer
  - The computer shares the printer
  - Computer needs to be running
- Print server
  - Print directly to the printer
  - Jobs are queued on the printer
  - Jobs are managed on the printer
  - Web-based front-end
  - Client utility

Configuration settings
- Duplex
  - Save paper
  - Print on both sides of the page without manually flipping over the paper
  - Not all printers can do this
- Orientation
  - Portrait vs. Landscape
  - The paper doesn't rotate
  - The printer compensates
- Tray settings
  - Printers can have multiple trays
  - Plain paper, letterhead, etc.
  - Choose the correct tray in the print dialog
- Quality
  - Resolution / Color, greyscale / Color saving

Printer security
- User authentication
  - Everyone can print
  - Set rights and permissions
  - Printing vs. managing the printer
- Badging
  - Authenticate when using the printer
  - Your job doesn't print until you use your employee badge
  - Quick and easy
- Audit logs
  - Cost management
  - Security monitoring
  - Event Viewer / System Events
  - May be built into the printer or print server
- Secured prints
  - Printer must support secure printing
  - Define a passcode
  - Use the passcode at the printer

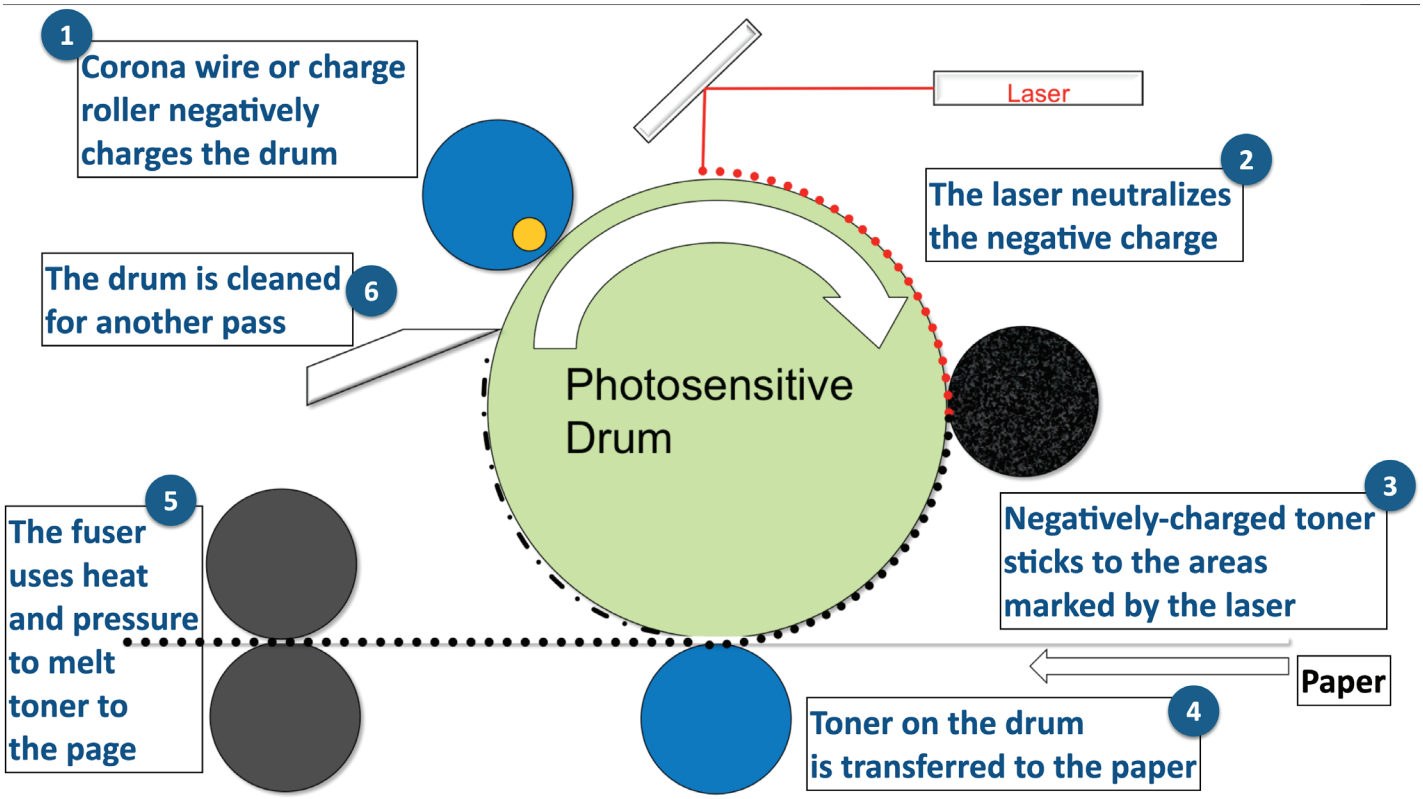
Flatbed scanner
- Different form factors
  - All-in-one multifunction device
  - Standalone flatbed
- May include an ADF
  - Automatic Document Feeder, multiple pages

Network scan services
- Scan to email
  - Scans are sent to your inbox
  - Large scans can fill up your mailbox
- Scan to folder
  - Using SMB (Server Message Block)
  - Send to a Microsoft share
- Scan to cloud
  - Cloud storage account
  - Google Drive, Dropbox, etc.

3.8 - Laser Printer Maintenance

Laser printer
- Combine a laser, high voltage, charged ions, powdered ink, heat, and paper
- Very high quality
- Fast printing speeds
- Very complex
  - Many moving parts
  - Requires on-printer memory
  - Messy on the inside

Replacing the toner cartridge
- Look for the messages
  - Low doesn't mean empty
- The toner can also contain the OPC drum
  - Organic Photoconductor drum
  - Sensitive to light; keep it in the bag
- Power down the printer
  - Safety first
- Remove packing strips from the new drum
  - Replace it with the old



Laser printer maintenance kit
- Laser printers wear out
  - All those moving parts
  - Heat and pressure
- Standard maintenance kits
  - Replacement feed rollers, new fuser unit, etc.
- When to perform maintenance?
  - Check the printer's page counter
- Power down and replace the components
  - Fuser units are HOT
- Reset the page counter when you're done!

Laser printer calibration
- Different toner cartridges print with different densities
  - Some dark, some light
- Laser printer calibration can adjust the density
  - Makes it looks perfect
- Can be automated or a manual process
  - Every printer is different
  - Check the printer manual

Laser printer cleaning
- Laser printers are dirty
  - All that toner and paper dust
- Check the manufacturer's recommendations
  - Water, isopropyl alcohol (IPA)
  - Don't use harsh chemicals!
- Outside - Damp cloth
- Inside - Wipe dust away
  - Don't use a normal vacuum cleaner or compressed air
  - Wash off skin with cold water
  - Clean rollers with IPA

3.8 - Inkjet Printers

Inkjet (ink-dispersion) printer
- Relatively inexpensive technology
- Quiet
- High-resolution
- Expensive ink
  – Proprietary
- Eventually fades
- Clogs easily

Ink cartridge
- Place drops of ink onto a page
  – Pulled from a set of cartridges
- CMYK - Cyan, Magenta, Yellow, Key (black)

Print head
- Some consumer printers integrate the print head into the ink cartridge
  – Change the cartridge, get a new print head
- Others separate the ink cartridge from the print head

Feed rollers
- Pick up and feed paper through the printer
  – Must be clean and not worn
- Duplexing
  – Print on both sides of the paper
  – Included with some printers

3.8 - Inkjet Printer Maintenance

Cleaning print heads
- Small droplets of ink - And small holes in a print head
- Clogged heads is a big issue
  – Many printers automatically clean every day
  – Output has streaks or sections of missing color
- Cleaning process can be started manually
  – Only takes a few minutes
- Some print heads/cartridges can be removed
  – Manually cleaning may help

Replacing inkjet cartridges
- Usually separate colors
  – Cyan, Magenta, Yellow, and Key (CMYK)
  – Some cartridges will combine these
- Takes seconds to replace
  – Takes a few minutes to calibrate and prepare the cartridge
- Recycle the empty cartridge - All plastic

Inkjet printer calibration
- Align nozzles to the paper
  – Lines should be crisp
  – Colors should not overlap
- Printer includes a calibration option
  – May need to make minor adjustments

Clearing jams
- Lots of turns and twists
  – A jam is inevitable
- Remove tray paper
  – Any loose paper
- Remove paper from the path
  – Firm pressure, don't rip
- Check for any scraps of paper
  - Remove all loose paper

3.8 - Thermal Printers

Thermal printer
- White paper
  – Turns black when heated
  – No ink!
- VERY quiet
  – Almost silent
- Paper is sensitive to light and heat
  – And clear tape

Feed assembly
- Pull paper through the printer
  – Relatively small paper path

Heating element
- Full-length heating element
  – No moving print head

Thermal paper
- Paper covered with a chemical
  – Changes color when heated
- Cash registers, credit card terminals
  – And quiet areas
- Looks like normal paper
  – Feels a bit different

3.8 - Thermal Printer Maintenance

Thermal paper replacement
- Relatively inexpensive
  - But impossible to substitute
- Different sizes
  - Not like laser printer paper - Keep a list
- Actual replacement process is easy
  - Simple paper feed
  - Small device

Cleaning the heating element
- Liquid cleaner
  - Isopropyl alcohol (IPA)
  - Get a cleaning pen
  - Check manufacturer's recommendations
- Swab gently - Usually small areas
- Use a cleaning card
  - Cleans the head and paper pathways

Removing debris
- Relatively small amount
  - Paper bits and dust
  - No toner!
- Blow out the printer - Take it outside
- Wipe it out - Damp cloth
- Avoid using a vacuum
  - Unless it's designed for computers
  - Resists static buildup/discharge

Paper sensitivity to heat
- Heat is used to create the output on the thermal paper
  - More heat will darken the entire page
- Avoid hot areas
  - Car dashboard, radiator, hair dryers
- Receipt will also fade over time
  - Don't use as archival media

3.8 - Impact Printers

Dot-matrix (impact) printers
- Print head with a small matrix of pins
  - Presses against a ribbon to make a mark on paper
- Good for carbon/multiple copies
- Low cost per page
- NOISY
- Poor graphics
- Relatively niche use cases

Dot-matrix printer head
- Moves back and forth
  - Pins hit ribbon and paper
- One matrix - Must move across the page to print

Printer ribbon
- Fabric
  - One long ribbon - Never ending circle
- Easy to replace - Once single unit
- Proprietary size - Specific to printer model

Tractor feed
- Paper pulled through with holes on the side of the paper
  - Instead of using friction
- Continuous paper feed - Perforations between pages
- Holes have to line up perfectly
  - Tractor paper can be perforated to remove holes

Multipart paper
- Carbonless copying
  - Multiple copies of a single form in a single pass
- Impact with the top sheet is duplicated on pages underneath
  - Micro-encapsulated ink on the back of the first sheet
  - Clay surface reaction on the top of the second sheet
- Impact activates these inks, dyes, and clay
  - Can be a mild irritation with certain chemical combinations
- Easier and less expensive to just print another copy
  - May be difficult to find multipart paper in use
  - Or dot-matrix printers

3.8 - Impact Printer Maintenance

Printer ribbon replacement
- Single ribbon - Self-contained - One long circle
- Replace when ink becomes too light
  - Ink is eventually consumed
- Designed to be modular - Replace in less than a minute

Print head replacement
- Takes a lot of abuse - Directly hits the ribbon and paper
- Gets hot - Watch your fingers
  - Another modular part - Look for a release lever or bar
- Replace with the ribbon for the best effect
  - The output should look perfect

Replacing paper
- Not as easy as a laser printer
  - Paper must feed perfectly into holes
  - Tractor feed
- Forms must be positioned correctly
  - Text needs to fit a predefined space
- Paper must feed without constraint
  - Make sure nothing is in the way

4.1 - Virtualization Concepts

Virtualization
- One computer, many operating systems
  - macOS, Windows 11, Linux Ubuntu, all at the same time!
- Separate OS, independent CPU, memory, network, etc.
  - But really one computer
- Host-based virtualization
  - Your normal desktop plus others
- Standalone server that hosts virtual machines
  - Enterprise-level
- Been around since 1967
  - IBM mainframe virtualization

Sandboxing
- Isolated testing environment
  - No connection to the real world or production system
  - A technological safe space
- Virtualize development process
  - Try some code, break some code, nobody gets hurt
- Additional development features
  - Roll back to a previous snapshot
  - Run additional systems

Building the application
- Develop
  - Secure environment
  - Writing code
  - Developers test in their sandboxes
- Test
  - A separate virtual environment for testing
  - Still in the development stage
  - All of the pieces are put together
  - Does it all work?

Legacy software and operating systems
- Need to run different application versions on the same system
  - Run each application instance in a separate VM
- Application only runs on a previous OS version
  - Create a VM with the older operating system

Cross-platform virtualization
- Windows / macOS / Linux doesn't do everything
  - Each OS has strengths and weaknesses
- Run different operating systems at the same time
  - Move between each OS seamlessly
  - No rebooting
- Save time and resources - One physical computer

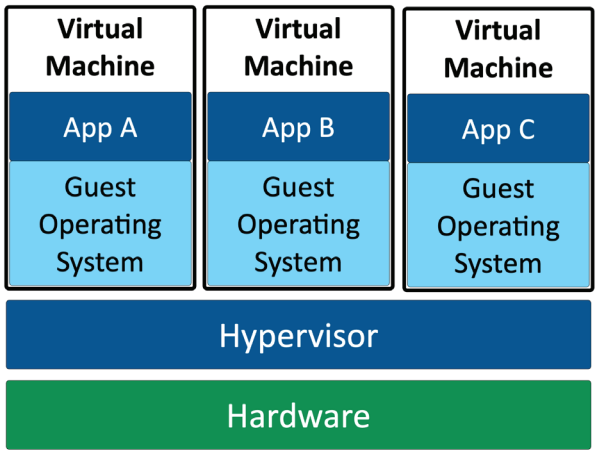
4.1 - Virtualization Services

The hypervisor
- Virtual Machine Manager
  - Manages the virtual platform and guest operating systems
- May require a CPU that supports virtualization
  - Can improve performance
- Hardware management - CPU, networking, security

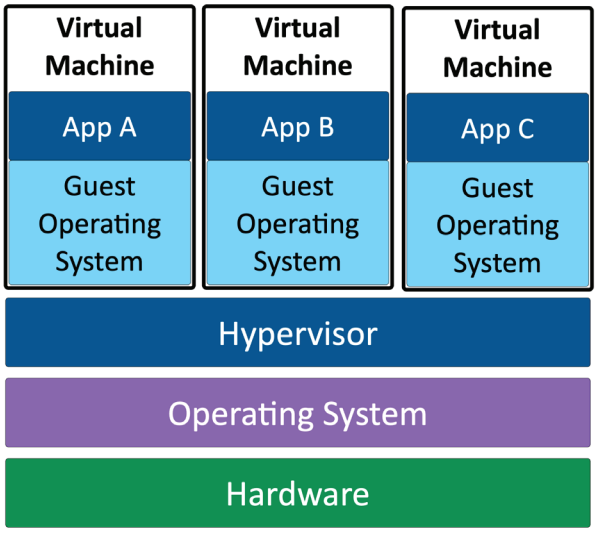
Virtualization
- Run many different operating systems on the same hardware
- Each application instance has its own operating system
  - Adds overhead and complexity
  - Virtualization is relatively expensive

Hypervisor types
- Type 1: Bare metal
  - The hypervisor is the primary operating system
  - VMware ESXi, Microsoft Hyper-V, Xen Project
- Type 2: Hosted
  - Hypervisor runs in the existing OS (Windows, Linux, macOS, etc.)
  - Virtual machines run on top of the current OS
  - VMware Workstation, Oracle VirtualBox, Parallels Desktop

Type 1 Hypervisor (Bare Metal)



Type 2 Hypervisor (Hosted)



4.1 - Virtualization Services (continued)

Resource requirements
- CPU Processor Support
  - Intel: Virtualization Technology (VT)
  - AMD: AMD-V
- Memory - Above and beyond host OS requirements
- Disk space - Each guest OS has it's own image
- Network
  - Configurable on each guest OS - Virtual switch

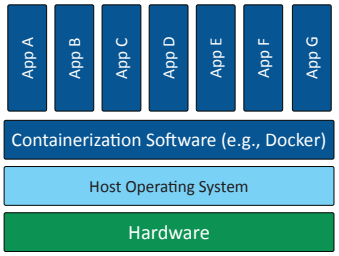
Network requirements
- Most client-side virtual machine managers have their own virtual (internal) networks
- Shared network address
  - The virtual machine shares the same IP address as the physical host
  - Uses a private IP address internally
  - Uses NAT to convert to the physical host IP
- Bridged network address
  - The VM is a device on the physical network
- Private address
  - The VM does not communicate outside of the virtual network

Hypervisor security
- Hypervisor is a sweet spot for the bad guys
  - No significant vulnerabilities yet
- VM escaping
  - Malware recognizes it's on a virtual machine
  - Malware compromises the hypervisor
  - Malware jumps from one guest OS to another
- Many hosted services are virtual environments
  - Malware on one customer's server can gather information from another

Guest operating system security
- Every guest is self-contained - Like a real computer
- Use traditional security controls
  - Host-based firewall, Anti-virus, anti-spyware
- Watch out for rogue virtual machines (VMs)
  - The bad guys try to install their own system
  - You're in big trouble
- Self-contained VMs provided by 3rd parties can be dangerous
  - You have no idea what's running on there

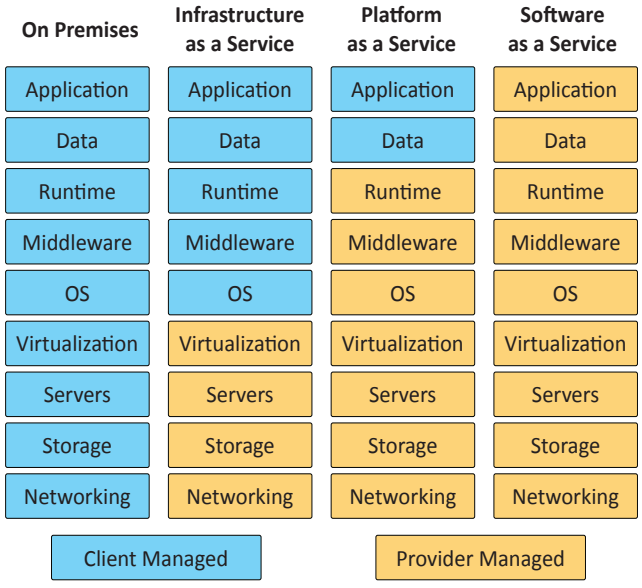
Virtual Desktop Infrastructure (VDI)
- Basic application usage
  - Applications actually run on a remote server
  - Desktop as a Service (DaaS)
- Minimal operating system on the client
  - No huge memory or CPU needs
- Network connectivity - Big network requirement
  - Everything happens across the wire

Application containerization
- Container
  - Contains everything you need to run an application
  - Code and dependencies
  - Standardized unit of software
- Isolated process in a sandbox
  - Self-contained
  - Apps can't interact with each other
- Container image
  - A standard for portability
  - Lightweight, uses the host kernel
  - Secure separation between applications



4.2 - Cloud Models

Cloud computing
- More than just a server hosted elsewhere
  - Extend the management of applications and services
- Effectively unlimited resources
  - You just have to pay for it
- Deploy an infrastructure in minutes
  - Create and tear down as you need
- International scope
  - Deploy in specific parts of the world
- Use only the required resources
  - Cost is based on the amount of use



4.2 - Cloud Models (continued)

Cloud deployment models
- Private - Your own virtualized local data center
- Public - Available to everyone over the Internet
- Hybrid - A mix of public and private
- Community - Organizations share the same resources

Infrastructure as a service (IaaS)
- Sometimes called Hardware as a Service (HaaS)
  - Outsource your equipment
- You're still responsible for the management
  - And for the security
- Your data is out there, but more within your control
  - Web server providers

Software as a service (SaaS)
- On-demand software
  - No local installation
  - Why manage your own email distribution? Or payroll?
- Central management of data and applications
  - Your data is out there
- A complete application offering
  - No development work required
  - Google Mail, Microsoft 365

Platform as a service (PaaS)
- No servers, no software, no maintenance team, no HVAC
  - Someone else handles the platform, you handle the development
- You don't have direct control of the data, people, or infrastructure
  - Trained security professionals are watching
  - Choose carefully
- Put the building blocks together
  - Develop your app from what's available on the platform
  - SalesForce.com

4.2 - Cloud Characteristics

Shared resources
- Internal/private cloud
  - Dedicated resources
  - Build your own cloud
  - Pay for everything up front
  - No ongoing costs
- External/public cloud
  - Share resources with a public cloud
  - Underlying infrastructure owned by a third-party
  - Cost may be metered or up-front

Metered and non-metered
- Metered utilization
  - You pay for what you use
  - Cost to upload - ingress traffic
  - Cost to store
  - Cost to download - egress traffic
- Non-metered utilization
  - You pay for a block of storage
  - No cost to upload
  - No cost to download

Cloud computing characteristics
- Elasticity
  - Seamlessly scale up and scale down as needed
  - The cloud enables instant resource provisioning
- Availability
  - Systems are always available
  - Redundancy provides availability
- File synchronization
  - Information can be duplicated across cloud locations
- Multitenancy
  - Many different clients are using the same cloud infrastructure

5.1 - Troubleshooting Hardware

POST (Power On Self Test)
- Test major system components before booting the operating system
  - Main systems (CPU, CMOS, etc.), video, memory
- Failures are usually noted with beeps and/or codes
  - BIOS versions can differ, check your documentation
- Don't bother memorizing the beep codes
  - They're all different between manufacturers
  - Know what to do when you hear them

POST and boot
- Blank screen on boot
  - Listen for beeps
  - Bad video, bad RAM, bad CPU, BIOS configuration issue
- BIOS time and setting
  - Maintained with the motherboard battery
  - Replace the battery
- Attempts to boot to incorrect device
  - Set boot order in BIOS configuration
  - Confirm that the startup device has a valid operating system
  - Check for media in a startup device

Crash screens
- Windows Stop Error or Blue Screen of Death
- Contains important information
  – Also written to event log
- Useful when tracking down problems
  – Sometimes more useful for manufacturer support

Bluescreens and spontaneous shutdowns
- Startup and shutdown BSOD
  – Bad hardware, bad drivers, bad application
- Use Last Known Good, System Restore, or Rollback Driver
  – Try Safe Mode
- Reseat or remove the hardware - If possible
- Run hardware diagnostics
  – Provided by the manufacturer
  – BIOS may have hardware diagnostics

Proprietary crash screens
- Every application has their own notifications
  – Some are very informational
  – Some are quite bad
- The quality of information contained in a help desk ticket can vary
  – Get as many details as possible
- A detailed transcript of the error would be valuable
  – Screenshots are even better
  – Ask for the screenshot in the ticket instructions

Blank screen
- Is the monitor connected?
  – We wouldn't ask if it wasn't a common solution
  – Check both power and signal cable
- Input selection on monitor - HDMI, DVI, VGA, etc.
- Image is dim - Check brightness controls
- Swap the monitor
  – Try the monitor on another computer
- No video after Windows loads
  – Use VGA mode (F8)

No power
- No power
  – No power at the source
  – No power from the power supply
  – Get out your multimeter
- Fans spin - no power to other devices
  – Where is your fan power connected?
  – No POST - bad motherboard?
  – Case fans have lower voltage requirements
  – Check the power supply output

Sluggish performance
- Task Manager
  – Check for high CPU utilization and I/O
- Windows Update - Latest patches and drivers
- Disk space
  – Check for available space and defrag
- Laptops may be using power-saving mode
  – Throttles the CPU
- Anti-virus and anti-malware
  – Scan for bad guys

Overheating
- Heat generation
  – CPUs, video adapters, memory
- Cooling systems
  – Fans and airflow, heat sinks, clean and clear
- Verify with monitoring software
  – Built into the BIOS
  – Try HWMonitor - http://www.cpuid.com/

Smoke and burning smell
- Electrical problems
  – The smoke makes everything work
- Always disconnect power
  – There should never be a burned odor
- Locate bad components
  – Even after the system has cooled down
  – Replace all damaged components

Random shutdown
- No warning, black screen
  – May have some details in your Event Viewer
- Heat-related issue
  – High CPU or graphics, gaming
  – Check all fans and heat sinks
  – BIOS may show fan status and temperatures
- Failing hardware
  – Has anything changed?
  – Check Device Manager, run diagnostics
- Could be anything
  – Eliminate what's working

Application crashes
- Application stops working
  – May provide an error message
  – May just disappear
- Check the Event Log
  – Often includes useful reconnaissance
- Check the Reliability Monitor
  – A history of application problems
  – Checks for resolutions
- Reinstall the application
  – Contact application support

5.1 - Troubleshooting Hardware (continued)

Unusual noises
- Computers should hum, not grind
- Rattling
  - Loose components
- Scraping
  - Hard drive issues
- Clicking
  - Fan problems
- Pop
  - Blown capacitor

Inaccurate system date/time
- Bad motherboard battery
  - Often a "button" style battery
- A bad battery will require a BIOS configuration or date/time configuration on every boot
- On older systems, can reset the BIOS configuration by removing the battery
  - Newer computers use a jumper

5.2 - Troubleshooting Storage Devices

Storage failure symptoms
- Read/write failure
  - "Cannot read from the source disk"
- Slow performance
  - Constant LED activity - Retry...retry...retry
- Loud clicking noise
  - The click of death
  - May also include grinding and scraping

Troubleshooting disk failures
- Get a backup - First thing - a bad drive is bad
- Check for loose or damaged cables
- Check for overheating
  - Especially if problems occur after startup
- Check power supply
  - Especially if new devices were added
- Run hard drive diagnostics
  - From the drive or computer manufacturer
  - Preferably on a known-good computer

Boot failure symptoms
- Drive not recognized, Boot Device Not Found
  - Lights (or no lights), beeps, error messages
- Operating system not found
  - The drive is there - Windows is not

Grinding noises
- Hard drives are mechanical devices
  - Spinning drives (often 5,400 RPM and higher)
  - Moving actuator arms
- Very high tolerances
  - One small problem can cause the drive to fail
- Metal on metal
  - Clicking or grinding
  - Poor performance, or no access at all
- Difficult to recover from this issue
  - Time to get your well-managed and very recent backup

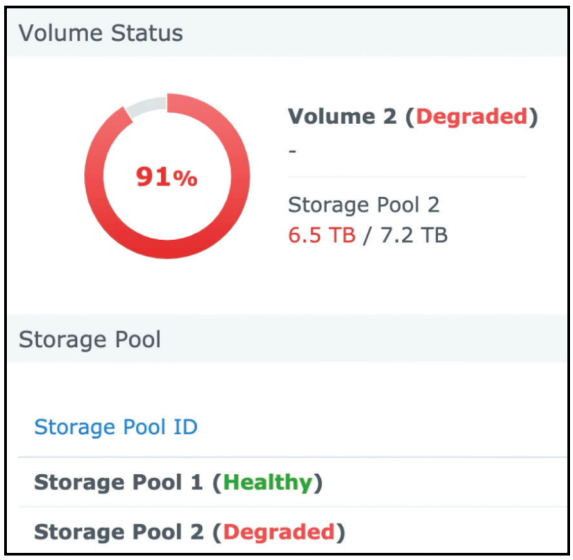
Troubleshooting disk failures
- Get a backup (if possible)
  - First thing - a bad drive is bad
- Check for loose or damaged cables
- Check for overheating
  - Especially if problems occur after startup
- Check power supply
  - Especially if new devices were added
- Run hard drive diagnostics
  - From the drive or computer manufacturer
  - Preferably on a known-good computer

Boot failure symptoms
- Drive not recognized, Boot Device Not Found
  - Lights (or no lights), beeps, error messages
- Operating system not found
  - The drive is there - Windows is not

Troubleshooting boot failures
- Check your cables - Physical problem
- Check boot sequence in BIOS
  - Check for removable disks (especially USB)
  - Check for disabled storage interfaces
- For new installation, check hardware configuration
  - Data and power cables
  - Try different SATA interfaces
- Try the drive in a different computer

Data loss/corruption
- Hard drives are mechanical devices
  - They will eventually fail
- Repairs are difficult and expensive
  - Dust-free environment
  - Not always successful
- An SSD may simply stop working
  - Sometimes can read but not write
- Data becomes unavailable or corrupted
  - Can be impossible to recover
- ALWAYS HAVE A BACKUP

RAID failure
- A drive in a RAID array has failed
  - Hardware failure, power issue, communication issue
- Almost always very obvious
  - Error messages
  - Email notifications
  - Audible alarms
- A careful analysis is required
  - Many drives, different volumes
- Each RAID is different
  - Don't start pulling drives until you check the console!



S.M.A.R.T.
- Self-Monitoring, Analysis, and Reporting Technology
  - Use third-party utilities
- Avoid hardware failure - Look for warning signs
- Schedule disk checks - Built-in to most drive arrays
- Warning signs - Replace a drive

S.M.A.R.T. analysis
- Monitoring S.M.A.R.T. metric over time
  - Watch for changes or incrementing values
  - May be part of a NAS or third-party software
- Automated notifications
  - Email
  - Text message
  - Console notifications
- Resolve the issue before it's a problem
  - Get a good backup
  - Replace the bad drive

Extended read/write times
- A lot happens when reading or writing data
  - Memory access, communication across the bus, spinning drive access, writing or reading the data to the storage device, etc.
- Delays can occur anywhere along the way
  - Need a way to measure storage device access
- Input/output operations per second (IOPS)
  - A broad metric of maximum performance
- Useful for comparing storage devices
  - Hard drive: 200 IOPS
  - SSD: 1,000,000 IOPS

Missing drives in OS
- OS boots normally
  - Other drives not shown - Check the BIOS
- Internal drives - Bad drive or disconnected cable
- External drives
  - No power to the drive or bad cable connection
- Network shares
  - Shared drives can be connected during startup
  - Option to reconnect at sign-in
  - Connected with login script

RAID recovery

RAID	Disks Required	Failure Condition
RAID 0	2 or more	A single drive failure breaks the array with data loss
RAID 1	2 or more	Array will work as long as one drive is operational
RAID 5	3 or more	Need all drives operational but one
RAID 6	4 or more	Need all drives operational but two
RAID 10 (RAID 1+0)	4 or more	Can lose all but one from each set of mirrors

5.3 - Troubleshooting Display Issues

Incorrect input source
- Is it connected?
  - We wouldn't ask if it wasn't a real solution
  - Check both power and signal cable
- Input selection on monitor
  - HDMI, DVI, VGA, etc.
- Image is dim
  - Check brightness controls
- Swap the monitor
  - Try the monitor on another computer
- No video after Windows loads
  - Use VGA mode (F8)

LCD projector bulbs
- LCD projectors and heat
  - Very bright metal-halide bulbs
  - Operate at around 1,000 degrees Celsius
- Cooling is always an issue
  - Fans constantly cool the bulb
  - Shuts down if temperatures are high
  - Fan cools bulb slowly after light is off
- Replace a burnt-out bulb
  - And clean dust from the air filters and external air input/output

Fuzzy image
- An LCD display is fixed
  - The number of pixels doesn't change
- A display looks best when the video settings match a display's native resolution
  - A mismatch can cause distortion
- If changing the display resolution,try using a multiple of the native resolution
  - 2560 x 1600 is the same ratio as 1920 x 1200

Burn-in
- A problem across all monitor types
  - Some displays will pixel-shift
  - You probably won't notice it
- LCDs have "image sticking"
  - Try to remove by displaying a white screen for an extended period

Dead pixels
- Always black
  - A manufacturing defect
- This probably isn't a problem on your side
  - Not a cable or power issue
- Clean the monitor screen
  - Damp cloth
- You'll have to replace the monitor
  - No other option

Flashing screen
- Check the video cable connections
  - A loose cable will cause loss of signal
- Replace the video cable
  - Usually a quick fix
- The problem could be with the monitor
  - Try switching the display
- Confirm the display settings in the operating system
  - May not match the monitor

Incorrect color display
- Colors may not be properly represented
  - Too much red, green, blue, yellow, etc.
- Monitor settings
  - Tint
  - Custom color presets
  - Factory Reset
- Driver configuration
  - Color tint
- OS configuration - Very blue
  - Night light settings

Audio issues
- Many monitors include speakers
  - No sound, low volume
- Monitor may have audio controls
  - Check volume levels and mute status
- Confirm audio input over HDMI, DisplayPort, or Thunderbolt
  - Check the HDMI audio output in the OS
- The monitor may support other audio input interfaces
  - Check for audio jacks

Dim image
- Monitor video settings
  - Brightness
  - Contrast
- Check the OS
  - Auto-dimming
  - Dim on battery power
  - Driver settings
- Backlight failure
  - Some or all of the light
  - May only be a section of the screen

5.3 - Troubleshooting Display Issues (continued)

Image quality problems
- Flickering, color patterns incorrect
  - You can almost work with this
- Check the cable pins
  - Especially if missing a color
- Distorted image and geometry
  - Check the OS refresh rate and resolution settings
  - Need to match the display specifications
  - Check or replace cable
- Disable hardware acceleration
  - Troubleshoot with the software drivers

Sizing issues
- The video output
  - The display may scale to fit the image (or not)
  - Check the display for a scaling option

Distorted image
- Lines, flashing colors, blocks
  - Lines, flashing colors, blocks
  - Difficult to see
- Hardware failure
  - LCD panels are very easy to break
  - No easy fix - replace the display
- Other hardware
  - Bad cable
  - Bad video adapter
- Software configuration
  - Bad video driver

5.4 - Troubleshooting Mobile Devices

Poor battery health
- Replace aging battery
  - There's only so many recharges
- Bad reception
  - Always searching for signal
  - Airplane mode on the ground
- Disable unnecessary features
  - 802.11 wireless, Bluetooth, GPS
- Check application battery usage
  - iOS and iPadOS: Settings/Battery
  - Android: Settings/Battery

Swollen battery
- Buildup of gas
  - Designed to self-contain
  - Do NOT open the battery packet/container
  - Significant fire risk
  - You'll be sorry
- Faulty battery
  - Stop using immediately
  - Dispose of properly
- Device can be damaged
  - Better than having a fire

Broken screen
- Time for a backup - Do this first!
- Replace the screen - No easy fixes
- The glass is sharp
  - Screen protectors can help
  - Use clear tape until repaired

Improper charging
- Check everything in the charging process
  - Cables, interfaces
- Check the cable interface
  - Remove any obstructions
- Check the cable
  - Don't use frayed or damaged cables
  - Try a known-good cable
- Verify the power adapter
  - Check with a multimeter

Poor or no connectivity
- Cellular or Wi-Fi
  - Location, location, location
- Cellular
  - Signal strength
  - Location matters
  - Outdoor connectivity may be better
- Wi-Fi
  - Limited range
  - Interference can limit throughput
  - May need a channel/frequency configuration change

Liquid damage
- Many phones have a Liquid Contact Indicator (LCI)
  - Turns color when liquid has been present
- Power down the phone
  - Don't power it back on. Don't do it.
- Remove the case, cards, back, and battery
  - If possible
- Putting the phone in a bag of rice doesn't actually work very well
  - Silica gel is better
- Leave it alone
  - Don't turn it on
  - Do not charge the phone
  - Don't connect it to power
  - Don't press any keys
  - Don't heat the phone
  - Don't move it around

- Wait at least a day
  - Or until the phone appears to be dry
  - And then perhaps a few hours after that
- Once you're sure, power it on
  - Hope for the best

Overheating
- Phone will automatically shut down
  - Avoid overheating damage
- Charging/discharging the battery, CPU usage, display light
  - All of these create heat
- Check app usage
  - Some apps can use a lot of CPU
- Avoid direct sunlight
  - Quickly overheats

Digitizer issues
- Touchscreen completely black or not responding to input
  - Screen presses do not register
- Apple iOS reset
  - Press power button, slide to power off, press power button
  - Hold down power button and Home|Volume for 10 seconds
- Android device restart
  - Remove battery, put back in, power on
  - Hold down power and volume down until restart
  - Some phones have different key combinations
  - Some phones do not have a key-based reset

Physically damaged ports
- External interface
  - Charging and data transfer
- Rough handling and accidents
  - Damages the interface
- Not charging
  - Look closely
- Not modular
  - Requires a system board replacement

Malware
- Always a concern
  - Even on mobile devices
  - The device type doesn't matter
- Look for symptoms
  - Unusual apps
  - Large data transfers
  - Pop up messages
  - High CPU usage, device overheating
  - Excessive battery use
- Try a security app or scanner
  - Identify malware infections

Cursor drift
- Random input or cursor moves without touching the device
  - Common on older screens
- Use a touch calibration app
  - Requires user input
  - Touch certain areas of the screen
- The system will adjust to the new calibration parameters
  - No more drift

Unable to install new applications
- Available space
  - Check storage capacity and delete unnecessary data
- Network connectivity
  - Apps are downloaded from an external resource
- OS or app compatibility
  - Upgrade the operating system and check for an updated application version
- Store cache or user credentials
  - Clear the Google Play Store cache or check iCloud username and password
  - Logout and log back in

Stylus does not work
- Power
  - Check the stylus battery charge and recharge if necessary
- Bluetooth connectivity
  - Check the connection and try re-pairing
- Stylus hardware issue
  - A stylus can suffer through a lot of physical wear
  - Check the tip and see if a tip replacement would apply
- Restart the device
  - Resets the operating system, the network, and everything else

Degraded performance
- Outdated OS and app software
  - Time to upgrade and restart
- Low storage
  - Slowdowns and crashes
- Hardware issues
  - Bad RAM or storage drive
- Excessive background apps
  - Excessive memory usage - try closing some apps
- Outdated hardware
  - New apps may not support older hardware

5.5 - Troubleshooting Networks

No network connectivity
- Do you have a link light?
  - Is it plugged in?
- Ping loopback (127.0.0.1)
  - Is the protocol stack working?
  - Availability and intermittent connectivity
- Ping local IP address
  - Checks local configuration, adapter, and link signal
- Ping default gateway
  - Connectivity on the local network
- Ping devices on router's other side
  - 8.8.8.8 or 9.9.9.9

Intermittent wireless connectivity
- Interference
  - Something else is using our frequency
- Signal strength
  - Transmitting signal, transmitting antenna, receiving antenna, etc.
- Incorrect channel
  - Usually automatic; look for manual tuning
- Bounce and latency
  - Multipath interference; flat surfaces
- Incorrect access point placement
  - Locate close to the users

Slow network speeds
- The network is slow!
  - It's probably not the network
- Confirm end-to-end connectivity
  - Ping or application login
  - Validate with a speed test
- Evaluate connectivity on each network hop
  - Utilization, errors, total throughput, filtering/ACLs, etc.
- May require a packet capture
  - The ultimate verification

Limited or no connectivity
- Windows alert in the system tray
  - "Limited or No connectivity"
  - "No Internet Access"
- Check the local IP address
  - An APIPA address will only have local connectivity
- If DHCP address is obtained,perform the ping tests
  - Local gateway, remote IP address

Jitter
- Most real-time media is sensitive to delay
  - Data should arrive at regular intervals
  - Voice communication, live video
- If you miss a packet, there's no retransmission
  - There's no time to "rewind" your phone call
- Jitter is the time between frames
  - Excessive jitter can cause you to miss information, "choppy" voice calls

Poor VoIP quality
- High speed and low latency
  - Real-time applications are demanding
- Check the Internet connection
  - A speed test can identify slow links
- Verify the local networking equipment
  - An old router can cause significant problems
- View the network performance
  - A packet capture would be useful

Port flapping
- Network interface goes up and down
  - Over and over again
- Verify the cable
  - Check the wiring
- Move between switch interfaces
  - Is the flapping associated with the switch interface or the device?
- Replace bad hardware or cables
  - May require additional purchases

High latency
- A delay between the request and the response
  - Waiting time
- Some latency is expected and normal
  - Laws of physics apply
- Examine the response times at every step along the way
  - This may require multiple measurement tools
- Packet captures can provide detailed analysis
  - Microsecond granularity
  - Get captures from both sides

External interference
- Predictable
  - Fluorescent lights
  - Microwave ovens
  - Cordless telephones
  - High-power sources
- Unpredictable
  - Multi-tenant building
- Measurements
  - Signal to noise ratio (SNR)
  - Performance Monitor

5.5 - Troubleshooting Networks (continued)

Signal to noise ratio (SNR)
- Signal
  - What you want
- Noise
  - What you don't want
  - Interference from other networks and devices
- You want a very large ratio
  - The same amount of signal to noise (1:1) would be bad

Authentication issues
- Access a resource
  - Requires the proper credentials
  - Username, password, other factors
- Verify an active session
  - May need to be refreshed
  - Logout and back in
- May be difficult to see
  - Part of a service or background process
- Perform a packet capture
  - Verify connectivity and look for errors

Intermittent Internet connectivity
- Determine the scope of the outage
  - Ongoing pings
  - Traceroute to a known location
  - Speed tests
- External issues require external support
  - Work directly with the Internet service provider (ISP)
  - Have your contact and account information available
- Check your SLA
  - Service level agreement

5.6 - Troubleshooting Printers

Testing the printer
- Print or scan a test page
  - Built into Windows, not the application
- Use diagnostic tools
  - Web-based utilities built into the printer
  - Vendor specific
  - Generic utilities

Bad output
- Lines down the printed page
  - Inkjet: Clean print heads
  - Laser: Check for scratched photosensitive drum
- Faded prints, blank pages - Low toner or ink
- Double/echo images or speckling
  - Laser printer optical drum not cleaned properly
  - Ghost or "shadow" from previous drum rotation

Garbled print
- Bad printer driver / wrong model
  - Incorrect page description language (PCL or PostScript?)
- Verify the printer functionality
  - Check with a test page
- Bad application
  - Check the output
  - Upgrade the application

Paper jam
- Careful when removing
  - Don't rip the paper
  - Don't damage internal components
- Paper not feeding or misfeeds multiple pages
  - Check the tray
  - Pickup rollers are part of a laser printer maintenance kit
- Creased paper
  - Problems in the paper path - Check the paper weight

Multiple prints pending in queue
- Corrupted print jobs
  - Print spooler will crash
  - Most spooler configuration will automatically restart
- Problems are logged
  - Windows Event Viewer, Windows-PrintService
- One job may be causing the issue
  - Monitor the queue for details

Grinding noises
- Never a good sound from your printer
  - The ink is already manufactured
- Something is not operating properly
  - Paper jam
  - Carriage is stalled or jammed
- Each printer has a different process
  - Check the manuals for removing paper jams
  - Some inkjet printers have a particular process
- May require additional maintenance
  - Or replace the printer

5.6 - Troubleshooting Printers (continued)

Finishing issues

- Finishing happens after the ink or toner has been applied
  - Collate, binding, etc.
- Staple jams
  - Larger printers will collate and staple groups of papers
  - Each manufacturer has a different removal process
- Incorrect hole punch location
  - Verify with the printer driver
  - May require a driver update

Incorrect page orientation

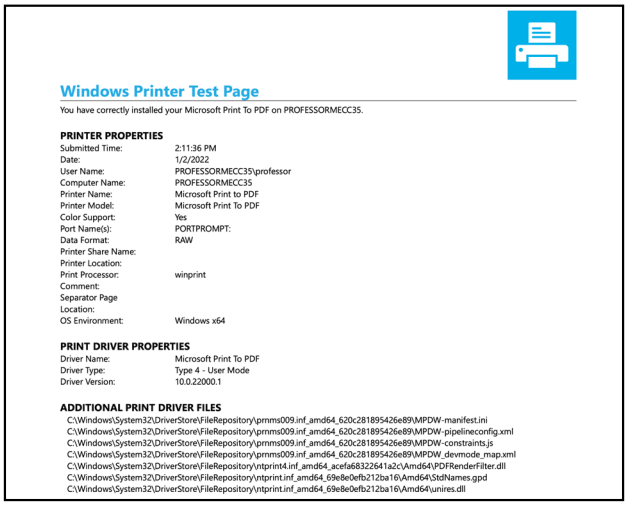
- Print a portrait page - Get a landscape print
- Check the settings when printing
  - Controlled by the print driver
  - May require a driver update
- The printer also has a default setting
  - Check the settings on the printer console

Tray not recognized

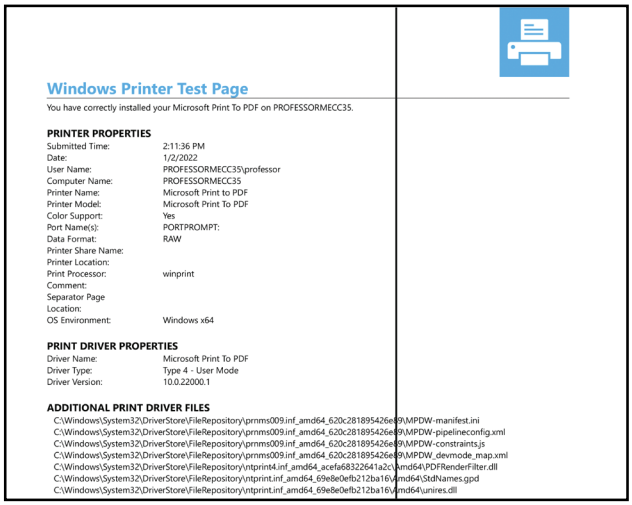
- Office printers often have multiple paper trays
  - Plain letter, plain legal, letterhead, paper colors, etc.
- The size of the printed page needs to match the size of the paper in the tray
  - You can't print a 14" legal page to an 11" letter page
- The definition of the paper tray is part of the driver configuration
  - Compare the user's configuration with the installed printer trays

Connectivity issues

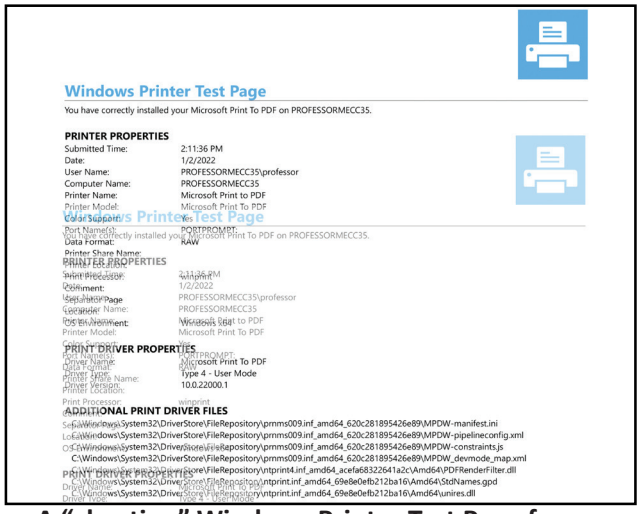
- The printer is a network device
  - Follows the same rules as any other device
- Confirm the connection type
  - Wired, wireless, etc.
- Verify IP address configurations
  - IP, subnet mask, default gateway, DNS, etc.
- Check the print server - May be part of the printer



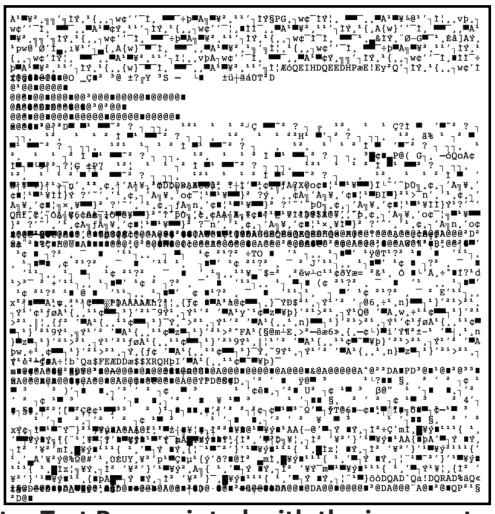
Normal Windows Printer Test Page



Windows Printer Test Page from a laser printer with a scratched photosensitive drum



A "ghosting" Windows Printer Test Page from a laser printer with a faulty cleaning subsystem



Printer Test Page printed with the incorrect page description language or incorrect driver

Continue your journey on
ProfessorMesser.com:

Professor Messer's Free
CompTIA A+ Training Courses

Monthly A+ Study Group Live Streams

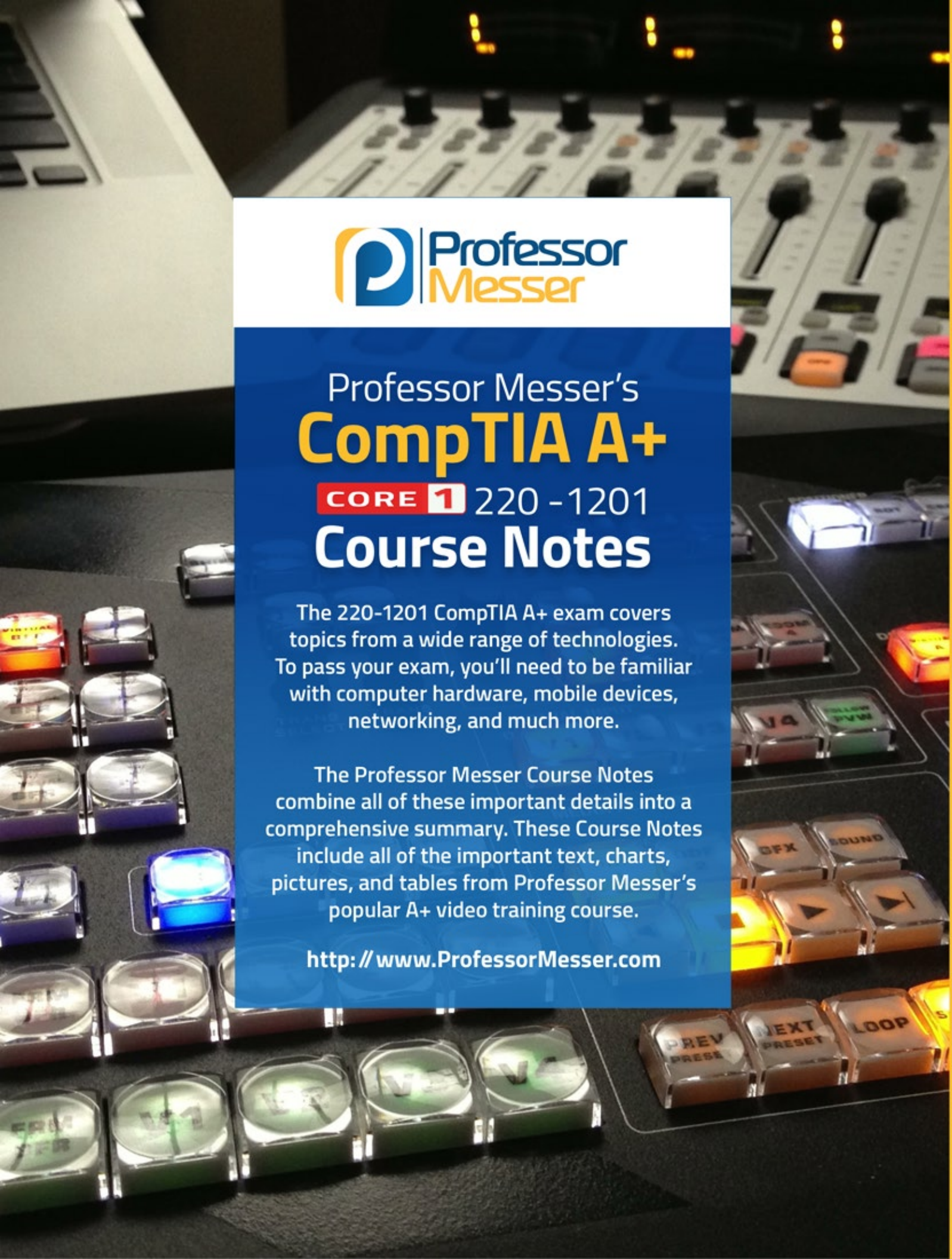
24 x 7 Live Discord Chat

Professor Messer's
CompTIA A+ Success Bundle

Voucher Discounts

Professor Messer

https://www.ProfessorMesser.com



Professor Messer's

CompTIA A+

CORE 1 220-1201

Course Notes

The 220-1201 CompTIA A+ exam covers topics from a wide range of technologies. To pass your exam, you'll need to be familiar with computer hardware, mobile devices, networking, and much more.

The Professor Messer Course Notes combine all of these important details into a comprehensive summary. These Course Notes include all of the important text, charts, pictures, and tables from Professor Messer's popular A+ video training course.

http://www.ProfessorMesser.com